

PROJECT REPORT ON

Machine Learning-Based Breast Cancer Survival
Risk Analysis Using Clinical and RNA-Seq Data
from TCGA-BRCA

COMPUTATIONAL CANCER BIOLOGY & BIOINFORMATICS

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1. INTRODUCTION

1.1 Background

Breast cancer is the most frequently diagnosed cancer among women worldwide and remains one of the leading causes of cancer-related mortality. According to the Global Cancer Observatory (GLOBOCAN 2022), breast cancer continues to impose a substantial global health burden despite significant improvements in cancer screening, diagnosis, and therapeutic interventions (Bray et al., 2024). Advances in surgical techniques, chemotherapy, radiotherapy, endocrine therapy, and targeted therapies have improved survival for many patients; however, breast cancer remains a major cause of morbidity and mortality because of its complex biological characteristics (Waks & Winer, 2019).

Breast cancer is a highly heterogeneous disease characterized by considerable variation in its molecular features, histopathological characteristics, clinical presentation, therapeutic response, and patient outcomes (Harbeck et al., 2019). This heterogeneity contributes to differences in disease progression among patients, even when they present with similar clinical characteristics. Consequently, predicting the clinical course of breast cancer remains a major challenge in oncology.

Accurate prognostic assessment plays an important role in breast cancer management by supporting treatment planning, patient counselling, and clinical decision-making. Reliable survival prediction models enable clinicians to classify patients according to their risk of adverse outcomes and facilitate more individualized therapeutic strategies. As precision oncology continues to evolve, there is increasing interest in incorporating molecular information alongside clinical characteristics to improve prognostic accuracy and better understand the biological mechanisms underlying disease progression (Harbeck et al., 2019; Waks & Winer, 2019).

1.2 RNA Sequencing in Breast Cancer Research

Recent advances in high-throughput sequencing technologies have enabled comprehensive analysis of cancer-associated molecular alterations. Among these technologies, RNA sequencing (RNA-seq) has emerged as a powerful approach for quantifying genome-wide gene expression and investigating transcriptomic changes associated with cancer development and progression. Combined with large-scale clinical datasets, RNA-seq provides an important resource for developing computational models capable of improving survival prediction and supporting precision medicine research (Bray et al., 2024).

Next-generation sequencing (NGS) technologies have revolutionized cancer research by enabling the detailed analysis of genomic and transcriptomic changes. One of the most popular technologies for measuring gene expression across the entire transcriptome is RNA sequencing (or ARN-seq), which has become increasingly prevalent. Compared to conventional methods, RNA-seq is more sensitive, has a wider dynamic range, and can identify alternative splice variants and transcripts that are not present in traditional microarray-based methods.

RNA-seq has made breast cancer research more accessible by providing information on gene expression patterns that influence tumour development, progression, metastasis, and patient survival. By identifying genes and biological pathways that play a role in disease progression, transcriptomic profiling can help identify molecular signatures that could be useful for both prognostication and therapy (Cancer Genome Atlas Network, 2012).

1.3 The Cancer Genome Atlas and TCGA-BRCA

The Cancer Genome Atlas (TCGA) is a comprehensive cancer genomics resource that was created to use standardized multi-omics datasets to characterize the molecular architecture of various human cancers. The TCGA Breast Invasive Carcinoma (TCG-BRCA) cohort compiles clinical data and RNA-seq gene expression profiles from numerous breast cancer patients. To investigate the connection between gene expression patterns and patient outcomes, this cohort provides a valuable source of matched clinical and transcriptomic data (Tomczak et al, 2015).

By combining RNA-seq data with clinical characteristics, more comprehensive analysis of factors that impact breast cancer prognosis is possible. Combined datasets facilitate the development of computational models for forecasting survival and provide opportunities to investigate the molecular mechanisms driving disease progression.

The rapid growth of high-throughput sequencing technologies has generated large-scale genomic datasets containing thousands of gene expression variables for individual patients. Although these datasets provide valuable biological information, their high dimensionality presents significant challenges for conventional statistical analysis. Machine learning techniques have therefore become increasingly important for identifying informative features, reducing data complexity, and improving predictive performance in biomedical research (Kourou et al., 2015).

1.4 Machine Learning and Survival-Based Prognostic Modelling

In breast cancer research, machine learning algorithms have been widely applied to classify patients, identify prognostic biomarkers, and predict clinical outcomes using genomic and transcriptomic data. Compared with traditional analytical approaches, machine learning methods can efficiently analyse complex datasets and uncover hidden patterns that may contribute to disease progression and patient survival (Kourou et al., 2015).

Among the various statistical approaches used in survival analysis, the Cox proportional hazards model is one of the most employed methods for investigating the relationship between clinical outcomes and multiple variables. When combined with high-dimensional gene expression data, feature selection methods such as the Least Absolute Shrinkage and Selection Operator (LASSO) help reduce the number of variables by retaining only the most informative predictors while minimizing overfitting (Tibshirani, 1997; Simon et al., 2011).

The integration of machine learning with survival analysis has facilitated the development of prognostic models capable of stratifying patients into different risk groups based on their predicted survival outcomes. These computational approaches enable the identification of survival-associated genes from high-dimensional transcriptomic data and

support the construction of robust prognostic models for breast cancer. Such models contribute to a better understanding of factors associated with patient prognosis and provide a foundation for further biological and clinical investigations (Kourou et al., 2015; Simon et al., 2011).

1.5 Rationale

Despite considerable progress in breast cancer diagnosis and treatment, accurately predicting patient outcomes remains challenging because of the molecular and clinical heterogeneity of the disease. Conventional prognostic assessment based mainly on clinicopathological characteristics may not fully capture the underlying biological differences among patients. The availability of high-throughput RNA sequencing (RNA-seq) data together with comprehensive clinical information from The Cancer Genome Atlas (TCGA) provides an opportunity to investigate molecular determinants of prognosis using an integrated analytical approach (Cancer Genome Atlas Network, 2012; Tomczak et al., 2015).

Several studies have reported that combining transcriptomic profiles with clinical characteristics improves the identification of prognostic biomarkers and enhances survival prediction compared with analyses based on a single data source (Zhu et al., 2017; Yu et al., 2019). Furthermore, statistical and machine learning approaches, such as Cox proportional hazards regression and Least Absolute Shrinkage and Selection Operator (LASSO) regression, enable efficient selection of survival-associated genes from high-dimensional transcriptomic datasets while minimizing model complexity and reducing the risk of overfitting (Tibshirani, 1997; Simon et al., 2011).

Therefore, this study utilized TCGA-BRCA clinical data and RNA-seq gene expression profiles to develop a machine learning-based prognostic model for breast cancer. A systematic analytical workflow involving survival-related gene identification, prognostic risk model construction, model validation, differential expression analysis, Gene Ontology (GO) enrichment, Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis, protein–protein interaction (PPI) network analysis, and clinicopathological correlation analysis was performed to investigate the biological significance of survival-associated genes and their relationship with clinicopathological characteristics in breast cancer

1.6 Aim

This study aimed to develop and evaluate a machine learning-based prognostic model for estimating overall survival risk in breast cancer patients by integrating clinical data and RNA sequencing (RNA-seq) gene expression data from The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) cohort.

1.7 Objectives

General Objective

To develop a prognostic model for estimating overall survival risk and stratifying breast cancer patients using RNA-seq gene expression data and clinical characteristics from the TCGA-BRCA cohort.

Specific Objectives

1. To obtain clinical and RNA-seq gene expression data from breast cancer patients in the TCGA-BRCA cohort.
2. To preprocess and match the clinical and transcriptomic datasets at the patient level for survival analysis.
3. To identify survival-associated genes using univariate Cox regression, LASSO regression, and multivariate Cox regression analyses.
4. To construct a three-gene prognostic risk model and calculate an individual risk score for each breast cancer patient.
5. To evaluate the prognostic performance of the risk model using Kaplan–Meier survival analysis, time-dependent ROC analysis, and Cox regression.
6. To identify differentially expressed genes between the high-risk and low-risk patient groups.
7. To investigate the biological functions and pathways associated with the differentially expressed genes using Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses.
8. To construct a protein–protein interaction (PPI) network to explore functional relationships among the differentially expressed genes.
9. To assess the association between patient risk groups and clinicopathological characteristics of breast cancer.

2. Review of literature

2.1.1 Global Burden of Breast Cancer

Breast cancer represents a major global public health burden and is the most commonly diagnosed cancer among women worldwide. According to GLOBOCAN 2022 estimates, approximately 2.30 million new cases of breast cancer and 666,000 breast cancer-related deaths occurred globally in 2022. These figures highlight the continuing importance of breast cancer as a major cause of cancer-related morbidity and mortality among women (Bray et al., 2024).

The burden of breast cancer varies across geographical regions because of differences in population demographics, risk factors, socioeconomic conditions, screening practices, early detection, and access to treatment. Although advances in screening, diagnosis, and treatment have improved outcomes for many patients, breast cancer continues to contribute substantially to global cancer mortality (Harbeck et al., 2019).

The substantial burden of breast cancer, together with variation in disease behaviour and patient outcomes, has encouraged continued investigation of the biological factors associated with prognosis. Molecular profiling approaches provide an opportunity to examine tumour characteristics beyond conventional clinical information and to investigate molecular features associated with patient outcomes.

2.1.2 Molecular and Clinical Heterogeneity of Breast Cancer

Breast cancer is a heterogeneous disease with distinct molecular, histopathological, and clinical characteristics. Differences in gene expression, genomic alterations, hormone receptor status, and tumour characteristics contribute to variation in disease behaviour and patient outcomes (Cancer Genome Atlas Network, 2012; Harbeck et al., 2019).

Molecular characterization has demonstrated biologically distinct breast cancer groups with differences in disease progression, treatment response, and prognosis. Thus, molecular information can complement conventional clinical characteristics in investigating factors associated with patient outcomes.

2.1.3 Heterogeneity and Patient Outcomes

Breast cancer heterogeneity contributes to differences in tumour behaviour and clinical outcomes among patients. Molecularly distinct tumours can differ in disease progression, treatment response, and prognosis, meaning that patients with similar clinical characteristics may experience different disease courses (Cancer Genome Atlas Network, 2012; Harbeck et al., 2019).

Gene-expression profiles can provide complementary information about tumour biology and may help identify molecular features associated with disease progression and survival. This provides a rationale for investigating transcriptomic features as potential prognostic factors in breast cancer.

2.1.4 Need for Molecular Prognostic Approaches

The molecular heterogeneity of breast cancer and its association with differences in patient outcomes highlight the need to investigate molecular features associated with disease behaviour and prognosis. Although clinicopathological characteristics remain important for prognostic assessment, molecular profiling can provide complementary information on tumour biology.

The availability of high-throughput transcriptomic data enables systematic investigation of gene-expression patterns associated with patient outcomes. Combined with survival modelling and feature-selection approaches, these data can be used to identify a smaller set of genes with potential prognostic relevance and investigate their utility for molecular risk stratification.

2.2.1 Overview of RNA Sequencing

RNA sequencing (RNA-seq) is a next-generation sequencing-based approach used to characterize the transcriptome by sequencing RNA-derived molecules and quantifying their abundance. Unlike methods that rely on predefined probes, RNA-seq provides sequence-based information that can be used to measure the expression of a large number of genes simultaneously. The resulting transcriptomic profiles can provide information about the molecular state of cells and tissues under different biological conditions (Wang et al., 2009; Stark et al., 2019).

RNA-seq can be used to investigate gene-expression differences, transcript isoforms, alternative splicing, and other features of the transcriptome. In cancer research, the measurement of transcript abundance provides an opportunity to characterize molecular changes associated with tumour development and progression and to investigate relationships between gene-expression patterns and clinical outcomes (Wang et al., 2009; Stark et al., 2019).

The ability of RNA-seq to generate genome-wide gene-expression profiles has made it particularly useful for large-scale transcriptomic studies. When gene-expression data are combined with clinical and survival information, they can be analysed to identify molecular features associated with disease outcomes and to investigate their potential relevance to prognostic modelling.

2.2.2 Advantages of RNA Sequencing for Transcriptomic Profiling

RNA sequencing provides broad transcriptome coverage and enables simultaneous analysis of gene expression across thousands of genes. Unlike probe-based approaches, RNA-seq can detect transcripts without requiring predefined targets and provides a broad dynamic range for transcript quantification (Wang et al., 2009; Stark et al., 2019).

The resulting high-dimensional expression data can provide detailed information about transcriptional variation in tumour samples. However, the large number of measured genes relative to the number of patients creates challenges for prognostic modelling and increases the need for appropriate preprocessing, gene filtering, and feature-selection methods.

2.2.3 Applications of RNA Sequencing in Breast Cancer Research

RNA-seq has been widely applied in breast cancer research to characterize transcriptomic variation and investigate molecular features associated with tumour biology. Gene-expression profiling can be used to identify transcriptional patterns associated with different biological and clinical characteristics and to investigate genes and pathways potentially related to tumour behaviour (Cancer Genome Atlas Network, 2012; Stark et al., 2019).

When RNA-seq data are analysed together with clinical and survival information, they can also be used to investigate associations between gene-expression patterns and patient outcomes. This provides a basis for identifying candidate prognostic genes and developing transcriptome-based risk models in breast cancer.

2.2.4 RNA Sequencing for Prognostic Gene Identification

Transcriptome-wide RNA-seq data enable investigation of associations between gene-expression patterns and patient survival. When expression profiles are analysed together with clinical and survival information, individual genes can be evaluated for their potential prognostic relevance in breast cancer (Stark et al., 2019; Harbeck et al., 2019).

However, the high dimensionality of RNA-seq data, with many genes measured across a relatively smaller number of patients, can increase model complexity and the risk of overfitting. Statistical filtering and feature-selection methods are therefore important for identifying a smaller set of informative genes (Tibshirani, 1997; Simon et al., 2011).

Cox proportional hazards regression can be used to evaluate associations between gene expression and survival outcomes, while LASSO regularization can reduce the number of candidate predictors by shrinking less informative coefficients toward zero. Together, these approaches provide a framework for selecting survival-associated genes and developing a compact prognostic signature from high-dimensional transcriptomic data (Tibshirani, 1997; Simon et al., 2011).

2.3 The Cancer Genome Atlas and TCGA-BRCA

2.3.1 Overview of The Cancer Genome Atlas

The Cancer Genome Atlas (TCGA) is a large-scale cancer genomics initiative established to characterize the molecular alterations associated with different cancer types. By integrating molecular profiles with clinical information, TCGA has provided a valuable resource for investigating cancer biology and identifying molecular features associated with disease characteristics and patient outcomes (Cancer Genome Atlas Network, 2012).

For breast cancer, TCGA generated comprehensive molecular profiles that demonstrated substantial biological heterogeneity among breast tumours. These datasets have subsequently supported numerous computational studies investigating molecular characteristics and prognostic factors in breast cancer.

2.3.2 TCGA-BRCA Cohort

TCGA-BRCA represents the breast invasive carcinoma project within The Cancer Genome Atlas and provides molecular and clinical information for breast cancer research. The availability of transcriptomic data together with patient-level clinical and survival information makes the cohort suitable for investigating associations between gene-expression patterns and clinical outcomes.

In the present study, TCGA-BRCA clinical information and RNA-seq gene-expression data were used as the primary data sources for constructing the survival-analysis cohort. The datasets were processed and matched at the patient level to obtain samples with the required expression and survival information for subsequent prognostic analysis.

2.3.3 Clinical and Molecular Data Available from TCGA-BRCA

TCGA-BRCA provides complementary clinical and molecular information that can be analysed at the patient level. Clinical information includes patient and disease characteristics together with survival-related variables, while molecular datasets provide tumour gene-expression profiles. The availability of these data allows molecular features to be investigated in relation to survival outcomes and clinicopathological characteristics.

In the present study, clinical and RNA-seq data from TCGA-BRCA were matched at the patient level to construct the cohort for survival analysis. Clinical variables were used to characterize the study population and to assess the association between the prognostic risk groups and clinicopathological characteristics, while RNA-seq gene-expression data were used for the identification of survival-associated genes and construction of the three-gene prognostic risk model.

The integration of these complementary data sources therefore provided the basis for examining both the **molecular characteristics associated with survival** and their **relationship with clinical characteristics** in breast cancer patients.

2.3.4 Use of TCGA-BRCA in Prognostic Research

TCGA-BRCA has been widely used in breast cancer research to investigate relationships between molecular characteristics and patient outcomes. The availability of transcriptomic profiles together with clinical and survival information allows researchers to identify genes associated with prognosis and develop molecular risk-stratification models.

Previous studies have applied survival-analysis and feature-selection approaches to TCGA-BRCA data to develop gene-based prognostic signatures. These studies demonstrate the usefulness of TCGA-BRCA as a resource for investigating transcriptomic features associated with breast cancer survival and provide a methodological basis for the present analysis.

In the present study, TCGA-BRCA was used to investigate survival-associated genes and construct a **three-gene prognostic risk model**, followed by evaluation of its association with survival and clinicopathological characteristics.

2.4 Clinical and Transcriptomic Data in Prognostic Modelling.

Clinical characteristics provide important information about disease severity and patient prognosis, while transcriptomic data provide information about the molecular state of the tumour. These two types of information therefore describe different aspects of breast cancer and can be examined together to investigate factors associated with patient outcomes.

Previous studies have demonstrated the value of integrating molecular and clinical information for cancer prognostic assessment. For example, multi-omics analyses have shown that combining molecular measurements with clinical variables can provide additional prognostic information compared with clinical characteristics alone in several cancer types (Zhu et al., 2017). Breast cancer studies have similarly used genomic, transcriptomic, and clinical information to develop prognostic models and evaluate their performance in independent patient cohorts (Yu et al., 2019).

In the present study, clinical and RNA-seq datasets were matched at the patient level. RNA-seq data were used to identify survival-associated genes and construct the three-gene prognostic risk model, while clinical characteristics were used to characterize the cohort and examine their association with the resulting risk groups.

2.4.2 Clinical Variables in Prognostic Assessment

Clinical characteristics such as age, tumour stage, lymph-node involvement, and other disease-related features provide important information for assessing prognosis in breast cancer. These variables reflect aspects of disease extent and patient characteristics and are commonly considered when evaluating differences in clinical outcomes.

However, clinical characteristics describe primarily the **clinical presentation and extent of disease**, whereas transcriptomic data provide information about the **molecular characteristics of the tumour**. Examining these complementary sources of information can therefore help investigate whether molecular risk groups are associated with established clinicopathological features.

In the present study, clinical characteristics including **age, sex, race, AJCC stage, T stage, N stage, M stage, and vital status** were examined in relation to the prognostic risk groups. This analysis was used to characterize the clinical features associated with the three-gene risk model rather than to claim that these variables were incorporated into the three-gene risk-score equation.

2.4.3 Integration of Clinical and Transcriptomic Information in Prognostic Studies

Integrating clinical and transcriptomic information allows cancer prognosis to be investigated from both clinical and molecular perspectives. Clinical variables describe patient and disease characteristics, whereas transcriptomic profiles provide information on tumour gene-expression patterns. Their combined analysis can therefore provide complementary information for investigating factors associated with patient outcomes.

Previous studies have used molecular and clinical information for prognostic assessment, demonstrating the potential of molecular features to complement conventional clinical variables (Zhu et al., 2017; Yu et al., 2019). In the present study, clinical and RNA-seq datasets

were matched at the patient level. RNA-seq data were used to identify survival-associated genes and construct the three-gene prognostic risk model, while clinical characteristics were examined in relation to the resulting risk groups.

2.4.4 Survival Analysis in Cancer Research

Survival analysis is widely used in cancer research to evaluate time-to-event outcomes while accounting for censored observations. Overall survival is commonly used as a prognostic endpoint, with the event generally defined as death during the follow-up period. Survival analysis therefore provides a framework for investigating associations between molecular or clinical characteristics and patient outcomes.

The **Cox proportional hazards regression model** is commonly used to assess the association between predictor variables and survival outcomes. It estimates hazard ratios that describe the relative risk of the event associated with predictor variables while accounting for follow-up time (Cox, 1972).

2.4.5 Cox Proportional Hazards Regression in Prognostic Modelling

Cox proportional hazards regression is widely used in survival analysis to evaluate the association between predictor variables and time-to-event outcomes. The model estimates a hazard ratio (HR), which indicates the relative change in the hazard of an event associated with a predictor while accounting for follow-up time (Cox, 1972).

In prognostic studies, Cox regression can be applied to identify survival-associated genes and to determine whether selected molecular features remain associated with survival after accounting for other variables. In the present study, Cox regression was used at different stages of prognostic gene selection and model evaluation.

2.5 Survival Analysis and Prognostic Modelling

2.5.1 Survival Analysis in Cancer Research

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2.5.2 Cox Proportional Hazards Regression in Prognostic Modelling

Cox proportional hazards regression is a widely used survival-analysis method for evaluating the association between predictor variables and time-to-event outcomes. The model estimates

a **hazard ratio (HR)**, which describes the relative hazard of an event associated with a predictor while accounting for follow-up time (Cox, 1972).

In prognostic gene-expression studies, gene expression can be evaluated in relation to **survival time and event status**, where death represents the event and patients for whom the event is not observed during follow-up are treated as censored observations. Cox regression can therefore be used to identify genes associated with overall survival and estimate their association with the hazard of death. This approach has been applied in cancer studies to investigate the prognostic significance of gene-expression features.

In the present study, Cox proportional hazards regression was used during **survival-associated gene selection and prognostic model evaluation**, forming part of the workflow used to identify the genes included in the three-gene prognostic risk model.

2.5.3 LASSO-Based Feature Selection

LASSO (Least Absolute Shrinkage and Selection Operator) is a regularization method that can perform variable selection by shrinking regression coefficients toward zero. In high-dimensional datasets, such as RNA-seq data containing thousands of genes, LASSO can reduce the number of candidate predictors and help control model complexity (Tibshirani, 1996; Tibshirani, 1997).

For survival data, LASSO can be incorporated into the Cox proportional hazards model to select genes associated with survival while applying a penalty to the model coefficients. Genes with less informative contributions may have their coefficients reduced to zero, allowing a smaller set of candidate genes to be retained for subsequent model construction (Tibshirani, 1997; Simon et al., 2011).

In the present study, LASSO-Cox regression was applied after initial survival-associated gene screening to reduce the number of candidate genes and identify features for subsequent multivariate Cox regression and construction of the three-gene prognostic risk model.

2.5.4 Prognostic Risk Score and Patient Stratification

A prognostic risk score can be constructed by combining the expression levels of selected genes with their corresponding coefficients obtained from a multivariable survival model. This produces an individual risk score for each patient, allowing patients to be stratified according to their estimated survival risk.

In the present study, the expression levels of the three selected prognostic genes were combined with their Cox regression coefficients to calculate an individual risk score. Patients were subsequently classified into **high-risk and low-risk groups** based on the predefined risk-stratification approach used in the analysis.

Risk stratification provides a framework for comparing survival outcomes between groups and for subsequently investigating differences in gene expression and clinicopathological characteristics.

2.5.5 Kaplan–Meier Survival Analysis

Kaplan–Meier analysis is a non-parametric method used to estimate the probability of survival over time while accounting for censored observations. It provides a graphical representation of the survival experience of a study population and allows survival distributions between groups to be compared.

In prognostic studies, Kaplan–Meier analysis can be used to compare survival between patients classified into different risk groups. In the present study, Kaplan–Meier survival analysis was used to compare **overall survival between the high-risk and low-risk groups** generated from the three-gene prognostic risk model.

A **log-rank test** can be used to assess whether the observed survival distributions between the groups differ statistically significantly. The Kaplan–Meier analysis therefore provides an important component of prognostic model performance evaluation.

2.5.6 Time-Dependent ROC Analysis

Time-dependent receiver operating characteristic (ROC) analysis is used to evaluate the ability of a prognostic model to discriminate between patients with different survival outcomes at specified time points. Unlike conventional ROC analysis, time-dependent ROC methods account for the time-to-event nature of survival data and the presence of censored observations.

The **area under the ROC curve (AUC)** provides a measure of the model's discriminatory performance at a given time point, with higher AUC values generally indicating better discrimination. In the present study, time-dependent ROC analysis was used to evaluate the discriminatory performance of the three-gene prognostic risk model at selected survival time points.

2.6 Prognostic Gene Signatures in Breast Cancer

2.6.1 Gene-Based Prognostic Signatures

Gene-expression signatures have been investigated as tools for estimating prognosis in breast cancer by combining the expression patterns of multiple genes into a molecular score. Several clinically developed multigene assays have demonstrated that gene-expression information can provide prognostic information in addition to conventional clinicopathological characteristics in selected breast cancer populations (Kim & Paik, 2010; Kwa et al., 2017).

Research-based prognostic signatures have also been developed using transcriptomic datasets and statistical modelling approaches. However, the clinical usefulness of a newly developed signature depends on appropriate model development, reproducibility, and independent validation. Therefore, identification of genes associated with survival within a discovery cohort alone is not sufficient to establish clinical utility (Subramanian & Simon, 2010; Weigelt et al., 2012).

In the present study, a **three-gene prognostic signature** was developed from TCGA-BRCA RNA-seq data and evaluated for its association with overall survival and clinicopathological characteristics.

2.6.2 Machine Learning and Prognostic Gene Selection

Machine learning and statistical learning approaches can be applied to high-dimensional gene-expression data to identify a smaller set of features associated with patient outcomes. In survival analysis, regularized methods such as LASSO-Cox regression are useful for reducing the number of candidate genes and controlling model complexity (Tibshirani, 1997; Simon et al., 2011).

Combining initial survival screening with regularized feature selection and multivariable Cox regression provides a structured approach for developing compact prognostic signatures. Such approaches have been widely used in transcriptomic prognostic studies because they can reduce the number of predictors relative to the original high-dimensional gene-expression dataset.

In the present study, **univariate Cox regression, LASSO-Cox regression, and multivariate Cox regression** were sequentially applied to identify the genes included in the final three-gene prognostic model.

2.6.3 Three-Gene Signatures and Model Complexity

Compact gene signatures can reduce the complexity of prognostic models while retaining a limited number of molecular features associated with patient outcomes. Compared with signatures containing a large number of genes, smaller models may be easier to interpret and investigate; however, a reduced number of genes does not by itself establish superior predictive performance or clinical utility. Appropriate model evaluation and independent validation remain important for assessing the generalizability of prognostic signatures (Subramanian & Simon, 2010; Weigelt et al., 2012).

In the present study, a **three-gene prognostic signature** was developed from TCGA-BRCA RNA-seq data following sequential survival-based feature selection. The model was evaluated within the TCGA-BRCA cohort using survival and discrimination analyses.

2.6.4 Research Gap

Although numerous gene-expression-based prognostic signatures have been investigated in breast cancer, differences in feature-selection methods, gene numbers, patient cohorts, and validation strategies can affect their reproducibility and generalizability. Large transcriptomic datasets also present challenges related to high dimensionality and model complexity.

Therefore, there remains a need to investigate compact prognostic signatures using systematic survival-based feature selection and to evaluate their association with patient survival and clinicopathological characteristics. The present study addresses this by developing a **three-**

gene prognostic risk model using TCGA-BRCA RNA-seq data, followed by evaluation of its prognostic performance and clinical associations within the study cohort.

2.7 Differential Gene Expression and Functional Enrichment

2.7.1 Differential Gene Expression Analysis

Differential gene expression analysis is used to identify genes whose expression levels differ between predefined biological or clinical groups. In cancer research, comparing gene-expression profiles between groups can help identify molecular features associated with differences in tumour characteristics and patient outcomes.

In prognostic studies, differential expression analysis can be applied after risk stratification to investigate molecular differences between **high-risk and low-risk groups**. The resulting differentially expressed genes (DEGs) can provide additional information about biological processes and pathways associated with the observed risk groups.

In the present study, differential gene expression analysis was performed between the **high-risk and low-risk groups** generated from the three-gene prognostic risk model. The identified DEGs were subsequently used for **GO and KEGG enrichment analyses** to investigate their potential biological significance.

2.7.2 Gene Ontology and KEGG Enrichment Analysis

Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses are commonly used to interpret the biological significance of gene sets identified from transcriptomic studies. GO analysis categorizes genes according to biological processes, molecular functions, and cellular components, whereas KEGG analysis helps identify biological pathways represented within a selected gene set.

In prognostic studies, enrichment analysis of differentially expressed genes can help identify biological processes and pathways associated with differences between patient risk groups. In the present study, **GO and KEGG enrichment analyses were performed using the differentially expressed genes identified between the high-risk and low-risk groups** to investigate the biological functions and pathways associated with the prognostic risk groups.

2.7.3 Functional Interpretation of Risk-Associated Genes

Functional enrichment analysis can provide biological context for genes that differ between prognostic risk groups. GO and KEGG analyses help identify biological processes and pathways represented among the differentially expressed genes, allowing molecular differences between high-risk and low-risk groups to be interpreted in relation to tumour biology.

In the present study, the enriched biological processes and pathways identified from the differentially expressed genes were used to characterize the molecular features associated with

the three-gene prognostic risk groups. These findings provide **biological context for the observed risk stratification** but do not establish a causal relationship between the identified pathways and patient survival.

2.8 Protein–Protein Interaction (PPI) Network Analysis

Protein–protein interaction (PPI) network analysis can be used to investigate functional relationships among proteins encoded by a selected set of genes. In such networks, proteins are represented as nodes and their reported or predicted associations as edges, providing additional biological context for transcriptomic findings.

The **STRING database** integrates information from multiple sources to construct protein-association networks and can be used to investigate relationships among proteins encoded by differentially expressed genes (Szklarczyk et al., 2021). In the present study, STRING was used to construct a PPI network from the differentially expressed genes identified between the high-risk and low-risk groups. The resulting network was used to support the functional interpretation of genes associated with the prognostic risk groups.

2.9 Clinicopathological Characteristics and Prognostic Risk Groups

Clinicopathological characteristics provide important information about the clinical and disease-related features of breast cancer patients. Examining these characteristics across prognostic risk groups can help determine whether molecular risk stratification is associated with differences in patient and disease characteristics.

In prognostic studies, characteristics such as age, tumour stage, lymph-node status, and other clinical variables are commonly examined in relation to molecular risk groups. Such analyses can provide clinical context for transcriptome-based prognostic models and help characterize the patients classified into different risk categories.

In the present study, **age, sex, race, AJCC stage, T stage, N stage, M stage, and vital status** were compared between the high-risk and low-risk groups using appropriate statistical tests. These analyses were performed to investigate the association between the three-gene prognostic risk groups and selected clinicopathological characteristics.

2.10 Summary of the Literature Review

Breast cancer is a heterogeneous disease with substantial variation in patient outcomes. RNA-seq provides transcriptome-wide information that can be combined with clinical and survival data to investigate molecular features associated with prognosis.

TCGA-BRCA provides a valuable resource for such analyses. Survival-analysis and feature-selection methods, including Cox regression and LASSO, can be used to identify survival-associated genes and develop compact prognostic signatures. GO, KEGG, and PPI analyses can further provide biological context for molecular differences between risk groups.

Based on this rationale, the present study used **TCGA-BRCA clinical and RNA-seq data to develop and evaluate a three-gene prognostic risk model** for breast cancer.

3. Materials and Methods

3.1 Study Design

This study employed a retrospective computational analysis of breast cancer data from the **TCGA-BRCA cohort** (<https://portal.gdc.cancer.gov>). Clinical and RNA-seq gene-expression data were obtained from publicly available TCGA resources and processed for patient-level integration. The analytical workflow included clinical and transcriptomic data preprocessing, gene quality control and filtering, survival-based feature selection, construction of a three-gene prognostic risk model, and evaluation of its prognostic performance. Differential gene expression, functional enrichment, protein–protein interaction, and clinicopathological association analyses were subsequently performed to further characterize the identified risk groups.

3.2 Data Collection from TCGA-BRCA

Clinical and RNA-seq gene-expression data for **breast invasive carcinoma (TCGA-BRCA)** were obtained from the publicly available **National Cancer Institute Genomic Data Commons (GDC)**. [NCI Genomic Data Commons](https://portal.gdc.cancer.gov)

Clinical and RNA-seq gene-expression data for **breast invasive carcinoma (TCGA-BRCA)** were obtained from the publicly available **National Cancer Institute Genomic Data Commons (GDC)**. [NCI Genomic Data Commons](https://portal.gdc.cancer.gov)

The clinical and transcriptomic datasets were processed and matched at the patient level to obtain the study cohort for subsequent survival and prognostic analyses. The final analytical cohort comprised **1,030 patients**.

3.3 Clinical Data Processing

Clinical data were processed using Python to obtain the variables required for survival analysis and clinicopathological characterization. Values representing missing clinical information were converted to missing values, and relevant clinical variables, including age at diagnosis, days to death, and days to last follow-up, were converted to appropriate numeric formats.

Overall survival time was derived according to vital status. For patients recorded as **dead**, the number of days to death was used as the survival time, whereas for patients without a recorded death event, the number of days to last follow-up was used. An event indicator was subsequently assigned as **1 for death and 0 for censored observations**.

Clinical characteristics including **age, sex, race, AJCC stage, T stage, N stage, M stage, and vital status** were retained for subsequent clinicopathological analysis. The clinical and RNA-seq datasets were matched at the patient level to ensure correspondence between clinical information and gene-expression profiles.

3.4 Integration of Clinical and RNA-seq Data

The processed clinical and RNA-seq datasets were integrated at the patient level using unique case identifiers. RNA-seq sample identifiers were mapped to corresponding patient identifiers

using the GDC sample metadata. Following identifier harmonization, patients with matching clinical and RNA-seq information were identified, and the datasets were merged using an inner join to generate the final integrated dataset for downstream analysis.

The resulting integrated dataset was used for subsequent gene-expression preprocessing, survival-based feature selection, and prognostic model construction.

3.5 Data Quality Control

The integrated TCGA-BRCA dataset was assessed for data completeness, variable types, duplicate patient records, and the validity of survival information. Missing values and the distributions of key clinical variables were examined before downstream analysis. Patient identifiers were checked for duplicate records to ensure that each patient was represented appropriately in the analytical dataset.

Survival time was also examined for invalid negative values. Patients with negative survival times were excluded from the dataset, while observations with valid non-negative survival times were retained for subsequent survival analysis.

3.6 Exploratory Data Analysis and Data Visualization

Exploratory analysis was performed to assess the distribution and characteristics of key clinical variables in the quality-controlled TCGA-BRCA cohort. Categorical variables, including **sex** and **race**, were visualized using count plots, while the distribution of age at diagnosis was examined using **histograms and boxplots**.

Age at diagnosis was converted from days to years using a factor of 365.25 days per year for visualization. The generated figures were used to assess the overall distribution of the clinical characteristics before subsequent transcriptomic and prognostic analyses.

3.7 RNA-seq Data Preparation

The quality-controlled TCGA-BRCA dataset was separated into clinical and gene-expression components. Clinical variables required for survival and downstream clinicopathological analyses were retained separately, while the remaining patient-level variables corresponding to gene-expression data were used to construct the gene-expression matrix.

The gene-expression dataset was assessed for **missing values, infinite values, duplicate columns, and data types** before downstream machine-learning analyses. The resulting clinical and gene-expression datasets were saved separately for subsequent preprocessing and prognostic modelling.

3.9 Variance Filtering and Low-Variance Gene Removal

Following removal of constant-expression genes, low-variance genes were filtered using the [VarianceThreshold](#) method with a variance threshold of **0.25**. Genes with variance below the predefined threshold were excluded, while genes meeting the threshold were retained for subsequent survival-based feature selection.

The filtered gene-expression matrix was subsequently used for **univariate Cox regression analysis**.

3.10 Univariate Cox Proportional Hazards Regression

Univariate Cox proportional hazards regression was performed separately for each gene to evaluate its association with overall survival. For each analysis, **survival time** was used as the duration variable and **event status** was used to indicate death or censoring. Genes with missing values for the corresponding expression or survival variables were excluded from the respective analysis.

For each gene, the Cox model estimated the **hazard ratio (HR)**, **regression coefficient**, **95% confidence interval (CI)**, and **p-value**. Because multiple genes were evaluated simultaneously, p-values were adjusted using the **Benjamini–Hochberg false discovery rate (FDR) procedure**. Genes with **FDR < 0.05** were considered statistically significant and were retained as candidate survival-associated genes for subsequent feature-selection analysis

3.11 Feature Standardization

Before LASSO-Cox feature selection, gene-expression features were standardized using **z-score standardization with StandardScaler**. This centred the expression values around zero and scaled each gene according to its standard deviation, placing the predictors on a comparable scale.

Patient identifiers, survival time, and event status were retained separately from the predictor matrix. Clinical outcome variables were not included as gene-expression predictors. The standardized gene-expression matrix was subsequently used for LASSO-Cox feature selection.

3.12 LASSO-Cox Feature Selection

Genes showing significant associations with overall survival after **Benjamini–Hochberg correction** in the univariate Cox analysis were subjected to **LASSO-penalized Cox regression** for feature selection. The analysis was performed using CoxnetSurvivalAnalysis with an **L1 penalty (l1_ratio = 1.0)**.

The LASSO-Cox model was evaluated across a regularization path of **30 alpha values**. The optimal alpha was selected using **five-fold cross-validation**, with model performance assessed using the concordance index. The alpha producing the highest mean cross-validation concordance was selected for the final LASSO-Cox model. Genes with **non-zero coefficients at the selected alpha** were retained for subsequent multivariate Cox regression and construction of the final prognostic signature.

3.13 Multivariate Cox Regression and Prognostic Model Construction

Genes retained by LASSO-Cox regression were subsequently evaluated together using a **multivariate Cox proportional hazards model**. The expression values of the selected genes were used as predictors, with overall survival time and event status defining the survival outcome. Regression coefficients were estimated for each selected gene and used to construct the prognostic risk score.

For each patient, the risk score was calculated as the linear combination of gene-expression values and their corresponding multivariate Cox regression coefficients:

where represents the expression value of the selected gene and represents its corresponding multivariate Cox regression coefficient. The resulting risk scores were retained for subsequent risk stratification and survival analyses.

$$Risk\ Score = \sum_{i=1}^n (\beta_i \times X_i)$$

C-index evaluation

The discriminative performance of the fitted prognostic model was assessed using the **concordance index (C-index)** based on the observed survival time, event status, and model-derived risk scores. The C-index was used to evaluate the ability of the model to correctly rank patients according to their survival risk.

3.14 Risk Stratification and Kaplan–Meier Survival Analysis

Patients were stratified into high- and low-risk groups according to the median prognostic risk score. Patients with a risk score greater than or equal to the median were assigned to the **high-risk group**, whereas those with a risk score below the median were assigned to the **low-risk group**.

Overall survival was subsequently compared between the two risk groups using the **Kaplan–Meier method**. Survival curves were generated with **95% confidence intervals**, and differences in survival distributions between the high- and low-risk groups were evaluated using the **log-rank test**. The number of patients at risk at different time points was also displayed to facilitate interpretation of the survival curves.

3.14 Internal Validation and Concordance Index Evaluation

The discriminative performance of the final three-gene Cox prognostic model was evaluated using the **concordance index (C-index)**. The apparent C-index was initially calculated using the complete TCGA-BRCA cohort. To assess potential optimism in the apparent model performance, **1,000 bootstrap resamples** were generated using a fixed random seed. The three-gene Cox model was refitted within each bootstrap sample, and the difference between the bootstrap and original-cohort concordance estimates was used to estimate model optimism.

The optimism-corrected C-index was calculated by subtracting the mean bootstrap optimism from the apparent C-index.

In addition, **five-fold cross-validation** was performed using shuffled folds with a fixed random seed. For each fold, the Cox model was fitted using the training subset and risk scores were generated exclusively for the held-out subset. The resulting out-of-fold risk scores were combined across all five folds, and the overall cross-validated C-index was calculated to assess the model's internal predictive discrimination.

3.15 Time-Dependent ROC Analysis

Time-dependent receiver operating characteristic (ROC) analysis was performed using the **timeROC package in R** to evaluate the discriminative performance of the prognostic risk score at predefined survival time points. Overall survival time, event status, and the model-derived risk score were used as the time-to-event outcome, event indicator, and prognostic marker, respectively. The analysis was performed at **1, 3, and 5 years (365, 1095, and 1825 days)** using the marginal weighting method, with death treated as the event of interest. The area under the ROC curve (AUC) was calculated for each time point, and time-specific ROC curves were generated to visualize the discriminatory performance of the prognostic model.

3.16 Statistical Analysis

All statistical analyses were performed using appropriate statistical and survival-analysis packages in Python and R. Overall survival was defined using the recorded survival time and event status, with death coded as the event and surviving patients treated as censored observations. Univariate and multivariate Cox proportional hazards regression were used to evaluate associations between gene expression and overall survival. Multiple-testing correction for the univariate gene-level analyses was performed using the **Benjamini–Hochberg false discovery rate (FDR)** method, with **FDR < 0.05** considered statistically significant. Kaplan–Meier survival curves were compared using the **log-rank test**. Model discrimination was assessed using the **concordance index (C-index)** and time-dependent AUC at 1, 3, and 5 years. Statistical analyses were conducted using the corresponding computational packages described in the preceding methodological sections.

3.17 Differential Gene Expression Analysis

Differential gene-expression analysis was performed to characterize transcriptional differences between the high- and low-risk groups defined by the prognostic risk score. Patients with risk scores greater than or equal to the median were classified as high risk, whereas those with scores below the median were classified as low risk. Gene expression between the two groups was compared using **Welch's two-sample t-test**, which does not assume equal variances between groups. For each gene, mean expression in the high- and low-risk groups, log2 fold change, and corresponding p-value were calculated. P-values were adjusted for multiple testing using the **Benjamini–Hochberg false discovery rate (FDR)** procedure. Genes with **FDR < 0.05 and absolute log2 fold change > 1** were considered differentially expressed.

3.18 Gene Annotation and Protein-Coding DEG Identification

The identified differentially expressed genes were annotated using gene identifiers obtained from the RNA-seq expression matrix. [Ensembl](#) gene version suffixes were removed to facilitate identifier matching, and duplicate gene identifiers were removed before annotation. Gene names and gene biotypes were subsequently assigned to the DEG results. Genes annotated as **protein-coding** were retained for downstream protein-coding DEG analysis. Significant protein-coding DEGs were defined using the same criteria of **FDR < 0.05 and |log2FC| > 1**.

3.18 Gene Ontology Enrichment Analysis

Gene Ontology (GO) enrichment analysis was performed to functionally characterize the significant protein-coding differentially expressed genes identified between the high- and low-risk groups. Gene symbols from the significant protein-coding DEG set were used as input for enrichment analysis using the [Enrichr](#) implementation of the [gseapy](#) package with the human-specific **GO Biological Process 2023, GO Cellular Component 2023, and GO Molecular Function 2023** gene-set libraries.

Enrichment results were filtered using an **adjusted p-value < 0.05**. For visualization, the top 20 significant terms were selected for each GO category. Gene counts and gene ratios were calculated from the genes contributing to each enriched term, and publication-quality dot plots were generated to represent the enriched Biological Process, Cellular Component, and Molecular Function categories.

3.19 KEGG Pathway Enrichment Analysis

KEGG pathway enrichment analysis was performed to identify biological pathways significantly enriched among the significant protein-coding differentially expressed genes. Gene symbols from the significant protein-coding DEG set were used as input for enrichment analysis using the [Enrichr](#) implementation of the [gseapy](#) package, with the human-specific **KEGG 2021 Human** gene-set library. Enrichment results were filtered using an **adjusted p-value < 0.05** to identify statistically significant pathways. For visualization, the top 20 significant pathways were ranked according to their combined enrichment score, and gene counts, gene ratios, and adjusted p-values were calculated for visualization in a bubble plot.

3.20 Gene Set Enrichment Analysis

Gene Set Enrichment Analysis (GSEA) was performed to investigate biological pathways associated with the transcriptional differences between the high- and low-risk groups. The analysis was conducted using a **pre-ranked gene list** generated from protein-coding genes, with genes ranked according to their **log2 fold-change values**. Duplicate gene symbols and missing values were removed before ranking. Pre-ranked GSEA was performed using the **gseapy package** with the **MSigDB Hallmark 2020** gene-set collection. The analysis used **1,000 permutations**, with gene sets restricted to a minimum size of 15 and a maximum size of 500 genes. A fixed random seed of 42 was used to ensure reproducibility.

Enriched gene sets were considered significant when the **nominal p-value was < 0.05 and the false discovery rate (FDR) q-value was < 0.25**. Significant gene sets were subsequently ranked according to their normalized enrichment score (NES) for downstream interpretation and visualization.

3.21 Protein–Protein Interaction Network and Hub-Gene Analysis

Protein–protein interaction (PPI) network analysis was performed using the **STRING database** to investigate functional interactions among the **913 significant protein-coding differentially expressed genes** identified between the high- and low-risk groups. Human gene identifiers were submitted to STRING, and interactions with a **combined interaction score ≥ 0.700** were retained to construct a high-confidence PPI network. Proteins without detected interactions were excluded from the connected network. The resulting PPI network was imported into **Cytoscape** for network visualization and topological analysis. **Cytoscape cytoHubba** was used to identify and rank candidate hub genes using the **Maximal Clique Centrality (MCC)** algorithm. Genes with higher MCC scores were considered highly central nodes within the PPI network and were selected as candidate hub genes for subsequent network visualization and biological interpretation.

METHODOLOGY FLOW CHART

*Development and Validation of a Three-Gene Prognostic Signature in Breast Cancer
(TCGA-BRCA)*



1. DATA ACQUISITION

TCGA-BRCA Clinical Data

- **Source:** GDC Portal (TCGA)
- **Data type:** Clinical information (Survival time, status, age, sex, stage, etc.)



TCGA-BRCA RNA-seq Data

- **Source:** GDC Portal (TCGA)
- **Data type:** HTSeq – FPKM (Gene expression profiles)



2. DATA PREPROCESSING

Clinical Data Processing

- Extract overall survival time
- Define event status (1 = Death, 0 = Alive)
- Remove missing or incomplete records



RNA-seq Data Processing

- Remove genes with low expression
- Log2 transformation
- Remove batch effects (if any)
- Match gene IDs



3. PATIENT MATCHING & FINAL COHORT

- Match clinical and RNA-seq data by patient ID
- Exclude patients without survival or expression data
- Final cohort used for analysis (N = 1,010)



4. GENE EXPRESSION FILTERING

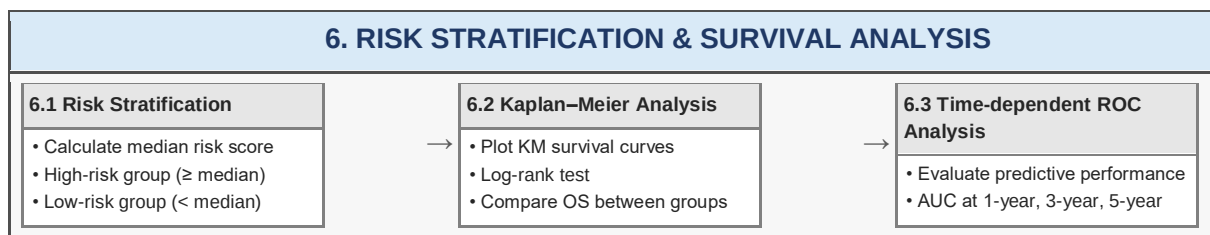
- Remove genes with low variance
- Keep most variable genes for downstream analysis

5. PROGNOSTIC GENE SELECTION & MODEL BUILDING

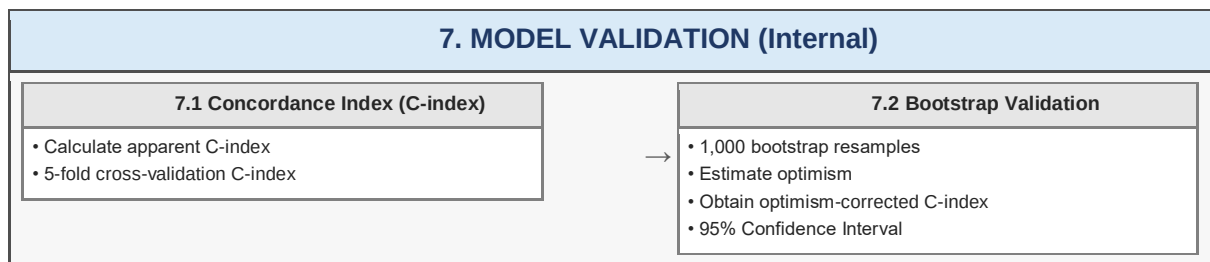
5.1 Univariate Cox Regression	5.2 Multiple Testing Correction	5.3 LASSO-Cox Regression	5.4 Multivariate Cox Regression	5.5 Risk Score Construction
- Perform Cox regression for each gene - Identify genes associated with OS ($p < 0.05$)	- Benjamini–Hochberg FDR correction - Select significant genes ($FDR < 0.05$)	- Feature selection using LASSO - Determine optimal λ using cross-validation - Select candidate genes	- Build multivariate Cox model with selected genes - Identify independent prognostic genes	Risk Score = $(\beta_1 \times \text{Gene}_1) + (\beta_2 \times \text{Gene}_2) + (\beta_3 \times \text{Gene}_3)$ (Three-gene signature)



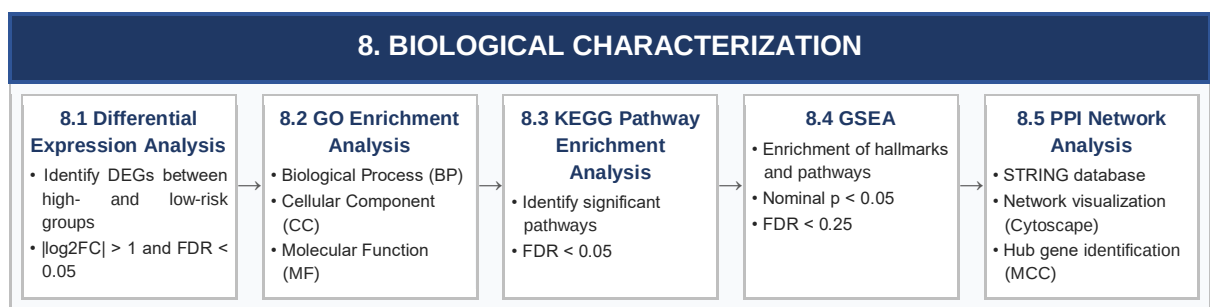
6. RISK STRATIFICATION & SURVIVAL ANALYSIS



7. MODEL VALIDATION (Internal)



8. BIOLOGICAL CHARACTERIZATION



9. INTERPRETATION & CONCLUSION

- Interpret the prognostic value of the three-gene signature
- Integrate survival analysis with functional and pathway insights
- Conclude the potential clinical relevance

5.Results

4.1 TCGA-BRCA Data Acquisition and Cohort Construction

TCGA-BRCA RNA-seq and clinical data were processed to construct a patient-level dataset for survival analysis. The downloaded RNA-seq files contained 60,664 rows corresponding to gene and auxiliary annotation records across the expression files. After processing and merging the RNA-seq samples, an expression matrix containing 1,232 samples was generated. The clinical dataset initially contained 1,098 cases. Patient identifiers were subsequently used to match the clinical and RNA-seq datasets, resulting in a matched cohort of 1,030 patients with both clinical survival information and RNA-seq expression data. The matched dataset was subsequently used for downstream quality control, feature filtering, survival analysis, and prognostic modelling.

Table 4.1. TCGA-BRCA cohort construction during data preprocessing

Processing stage	Samples / dimensions
Clinical dataset	1,098 patients
RNA-seq expression matrix	1,232 samples
Matched clinical/RNA-seq cohort	1,030 patients
Final matched expression matrix	1,030 × 47,719
Final merged dataset	1,030 × 47,728

5.2 Clinical Characteristics of the TCGA-BRCA Cohort

The TCGA-BRCA cohort comprised 1,030 patients and was predominantly female. The cohort showed a broad age distribution, with most patients diagnosed during middle to older adulthood. The majority of patients were White, followed by Black or African American and Asian patients. **Most patients had AJCC Stage IIA (332, 32.2%)** or Stage IIB (231, 22.4%), followed by Stage IIIA (138, 13.4%) and Stage IA (81, 7.9%). Stage I accounted for 80 patients (7.8%), while Stage IIIC, IIIB, and IV accounted for 56 (5.4%), 22 (2.1%), and 18 (1.7%) patients, respectively. Stage information was unavailable for 52 patients (5.0%).

For T stage, **T2 was the predominant category (578, 56.1%)**, followed by T1c (200, 19.4%) and T3 (117, 11.4%). The remaining patients were distributed among the less frequent T-stage subcategories, with 43 patients (4.2%) having unavailable information.

For N stage, **N0** was the most frequent category (**310, 30.1%**), followed by N1a (156, 15.1%), N0 (i-) (138, 13.4%), and N1 (107, 10.4%). Other N-stage categories occurred at lower frequencies, while 43 patients (4.2%) had unavailable N-stage information.

For **M stage**, the majority of patients were classified as **M0 (823, 79.9%)**. MX accounted for 140 patients (13.6%), while M1 was observed in 19 patients (1.8%). Five patients (0.5%) were classified as cM0 (i+), and 43 patients (4.2%) had unavailable M-stage information.

Overall, the cohort was characterized predominantly by female patients, Stage IIA/IIB disease, T2 primary tumor, N0 lymph-node status, and M0 metastatic status. Unreported or missing clinical values were retained as unknown/not available and were not assigned to a biological stage.

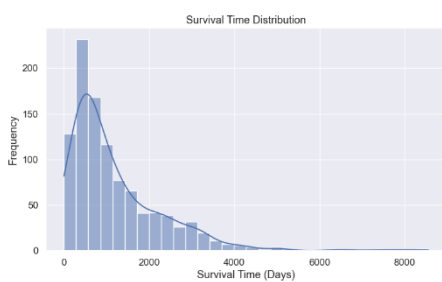


Figure 5.1: Survival-time distribution

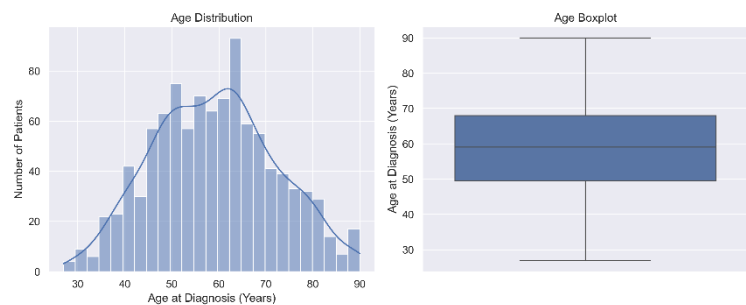


Figure 5.2: Age distribution

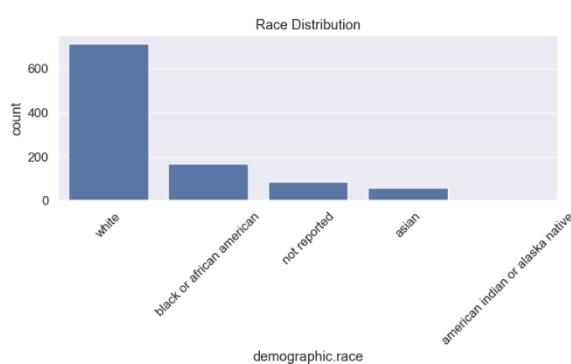


Figure 5.3: Race distribution

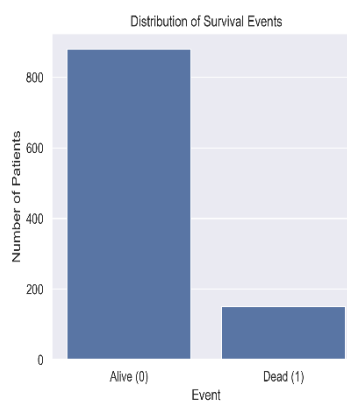


Figure 5.4: Survival-event distribution

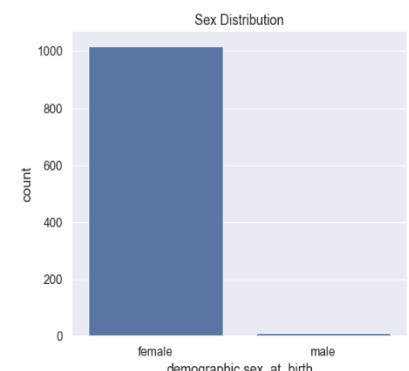
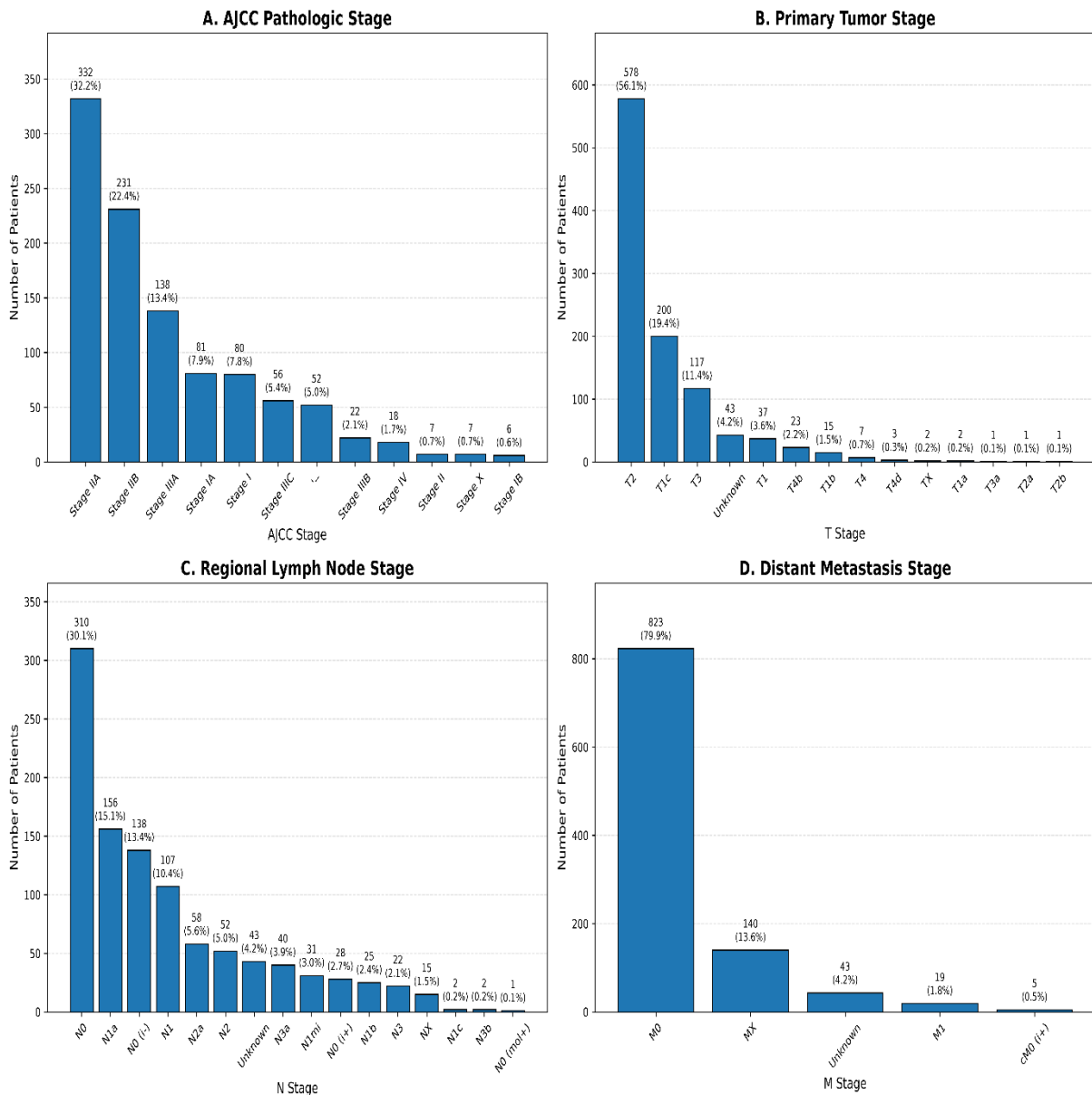


Figure 5.5: Sex distribution

Figure 5.2. Distribution of clinicopathological characteristics in the TCGA-BRCA cohort. The distribution of patients according to (A) AJCC pathological stage, (B) primary tumor (T) stage, (C) regional lymph-node (N) stage, and (D) distant metastasis (M) stage is shown. Numbers above the bars indicate the number and percentage of patients in each category. Unavailable or unreported values were classified separately as unknown.

Clinicopathological Characteristics of the TCGA-BRCA Cohort



5.3 Gene Expression Quality Control

Quality-control assessment was performed on the cleaned gene-expression matrix before feature selection. The dataset contained 1,030 patient samples and 57,914 gene-expression features. No missing values or infinite values were detected, and no duplicate gene names or constant-expression genes remained in the dataset. Expression values ranged from 0 to 7,032,374, with a mean expression value of 1,008.44 and a median of 1.00. These quality-control checks indicated that the processed expression matrix was suitable for subsequent variance-based feature filtering.

TCGA-BRCA Gene Expression Quality Control Report	

Input Information	
Input File	: gene_expression_no_constant.csv
Dataset Shape	: (1030, 57915)

Quality Control Checks	
Missing Values	: 0
Infinite Values	: 0
Duplicate Gene Names	: 0
Constant Genes	: 0

Expression Statistics	
Minimum Expression	: 0.00
Maximum Expression	: 7032374.00
Mean Expression	: 1008.44
Median Expression	: 1.00

Quality Control Status	
✓ Dataset Loaded Successfully	
✓ Missing Values Checked	
✓ Infinite Values Checked	
✓ Duplicate Gene Names Checked	
✓ Constant Genes Checked	
✓ Dataset Ready for Feature Selection	

Output Files	
gene_expression_no_constant.csv	
gene_qc_report.txt	

Figure 5.6. TCGA-BRCA gene-expression quality-control summary.

5.4 Low-Variance Gene Filtering

To reduce the number of low-information molecular features, variance-based filtering was performed using a variance threshold of **0.25**. The analysis was conducted on **1,030 patient samples**, and no patients were removed during this filtering step. Of the **57,914 gene features** evaluated, **47,718 genes** were retained, whereas **10,196 genes** were removed, corresponding to **17.61%** of the original gene features. Thus, the final filtered expression matrix contained **1,030 patients and 47,718 gene features** and was subsequently used for prognostic gene identification and survival modelling.

TCGA-BRCA Low Variance Filtering Summary	

Input Information	
Input File	: gene_expression_no_constant.csv
Variance Threshold	: 0.25

Dataset Summary	
Samples Before	: 1030
Samples After	: 1030
Genes Before	: 57914
Genes After	: 47718
Removed Genes	: 10196
Percent Removed	: 17.61%

Filtering Status	
✓ Dataset Loaded Successfully	
✓ Variance Calculated	
✓ Low Variance Genes Removed	
✓ Filtered Dataset Saved	
✓ Dataset Ready for Univariate Cox Regression	

Output Files	
gene_expression_variance_filtered.csv	
variance_filter_report.txt	

Figure 5.7 Summary of low-variance gene filtering

5.5 Univariate Cox Regression

Univariate Cox proportional hazards regression was performed for individual gene-expression features to evaluate their association with overall survival in the TCGA-BRCA cohort. A total of **684 genes** were identified as significantly associated with overall survival following Benjamini–Hochberg correction at an FDR threshold of 0.05. These genes were subsequently considered candidate prognostic features for LASSO-Cox feature selection.

The strongest associations included.

ENSG00000121871 (HR = 1.014, 95% CI: 1.013–1.015, FDR = 2.32×10^{-197}),

ENSG00000231436 (HR = 2.807, 95% CI: 2.611–3.019, FDR = 1.14×10^{-166}),

ENSG00000171951 (HR = 1.001, 95% CI: 1.001–1.001, FDR = 4.50×10^{-118}),

ENSG00000231421 (HR = 1.038, 95% CI: 1.035–1.042, FDR = 2.72×10^{-113}),

ENSG00000231358 (HR = 1.143, 95% CI: 1.129–1.157, FDR = 3.27×10^{-99}).

Table 5.3. Representative genes showing strong associations with overall survival in univariate Cox regression.

Gene	Hazard Ratio	95% CI	Coefficient	P-value	FDR
ENSG00000121871	1.014	1.013–1.015	0.0136	4.86×10^{-202}	2.32×10^{-197}
ENSG00000231436	2.807	2.611–3.019	1.0323	4.78×10^{-171}	1.14×10^{-166}
ENSG00000171951	1.001	1.001–1.001	0.0011	2.83×10^{-122}	4.50×10^{-118}
ENSG00000231421	1.038	1.035–1.042	0.0376	2.28×10^{-117}	2.72×10^{-113}
ENSG00000231358	1.143	1.129–1.157	0.1336	3.43×10^{-103}	3.27×10^{-99}

5.6 LASSO-Cox Feature Selection

Following univariate Cox regression and Benjamini–Hochberg false discovery rate correction, **684 genes** significantly associated with overall survival were retained as candidate prognostic features. These genes were subsequently subjected to LASSO-penalized Cox regression to reduce the number of candidate features and identify a smaller gene set for subsequent multivariate modelling. The LASSO-Cox analysis was performed using an **L1 penalty (l1_ratio = 1.0)** on a cohort of **1,030 patients**, including **150 observed survival events and 880 censored observations**.

The LASSO-Cox model was evaluated across a regularization path of **30 alpha values** using **five-fold cross-validation**, with the concordance index used to assess model discrimination. Among the alpha values for which valid cross-validation estimates were obtained, the highest mean cross-validated concordance index was **0.5271**, observed at an alpha value of **0.04637**. At the selected regularization parameter, three genes retained non-zero coefficients: **ENSG00000004846 (ABCB5)**, **ENSG00000124343 (XG)**, and **ENSG00000269495 (AC011483.2)**, with LASSO coefficients of **0.0955**, **0.0555**, and **0.0134**, respectively. These three genes were subsequently carried forward to multivariate Cox regression for construction of the final prognostic model.

Table 5.4. LASSO-Cox selected genes

Gene ID	Gene symbol	LASSO coefficient
ENSG00000004846	ABCB5	0.0955
ENSG00000124343	XG	0.0555
ENSG00000269495	AC011483.2	0.0134

5.7 Multivariate Cox Regression

The three genes selected through LASSO-Cox regression were subsequently evaluated using multivariate Cox proportional hazards regression to estimate their joint association with overall survival. The analysis was performed using the expression values of **ENSG00000004846 (ABCB5)**, **ENSG00000124343 (XG)**, and **ENSG00000269495 (AC011483.2)** in the complete cohort of **1,030 patients**. All three genes were retained in the multivariate model, with estimated regression coefficients of **0.1338**, **0.1211**, and **0.1201**, respectively. The positive coefficients indicate that higher expression values of each gene contributed positively to the model-derived prognostic risk score. The resulting multivariate Cox model was subsequently used to calculate an individual risk score for each patient and construct the three-gene prognostic signature.

Table 5.5. Multivariate Cox regression coefficients of the three-gene prognostic signature

Gene ID	Gene symbol	β coefficient
ENSG00000004846	ABCB5	0.1338
ENSG00000124343	XG	0.1211
ENSG00000269495	AC011483.2	0.1201

5.8 Three-Gene Prognostic Risk Score

Based on the multivariate Cox regression coefficients, an individual prognostic risk score was calculated for each patient as a linear combination of the standardized expression values of the three selected genes:

$$\text{Risk Score} = (0.1338 \times \text{ABCB5}) + (0.1211 \times \text{XG}) + (0.1201 \times \text{AC011483.2})$$

The calculated risk scores were assigned to the corresponding patients and subsequently used for risk stratification into high- and low-risk groups according to the median risk score. The model generated a risk score for all **1,030 patients**. The apparent concordance index of the three-gene Cox model was **0.5658**.

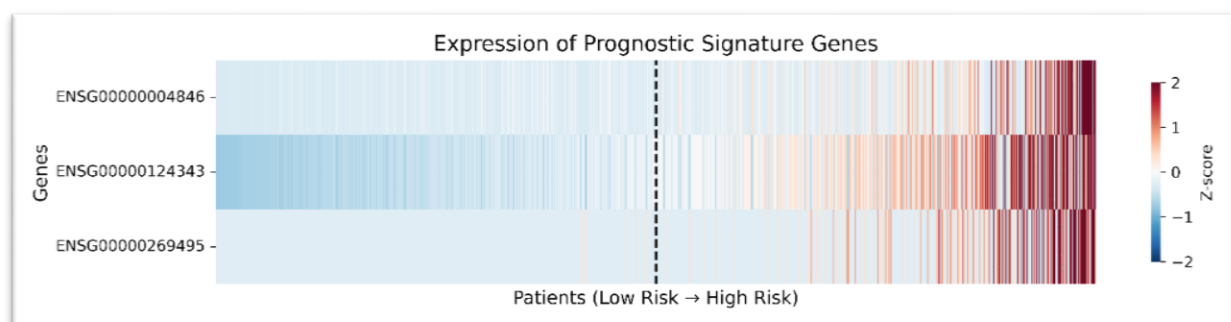


Figure 5.8 Expression patterns of the three-gene prognostic signature across TCGA-BRCA patients. Patients are ordered from low to high risk according to the multivariate Cox-derived risk score. Gene-expression values are represented as Z-scores. The dashed vertical line indicates the median risk-score cutoff separating the low- and high-risk groups.

5.9 Risk Stratification

The prognostic risk score was calculated for all **1,030 TCGA-BRCA patients** using the coefficients derived from the multivariate Cox regression model. The **median risk score was -0.0772** and was used as the cutoff to stratify patients into high- and low-risk groups. Patients with a risk score **greater than or equal to the median** were classified as high risk, whereas patients with a risk score below the median were classified as low risk.

The median-based stratification resulted in **515 patients in the high-risk group and 515 patients in the low-risk group**, providing equal group sizes for subsequent Kaplan–Meier survival analysis.

Table 5.6. Distribution of patients according to prognostic risk group

Risk group	Number of patients	Percentage
Low Risk	515	50.0%
High Risk	515	50.0%
Total	1,030	100%

5.10 Kaplan–Meier Survival Analysis

Kaplan–Meier analysis demonstrated a statistically significant difference in overall survival between the high- and low-risk groups (log-rank $p = 1.173 \times 10^{-3}$), indicating that the risk-score-based stratification was associated with different observed survival distributions. However, the time-dependent ROC analysis showed limited discriminatory performance at 1 year (AUC = 0.527) and 3 years (AUC = 0.549), with somewhat improved but still modest discrimination at 5 years (AUC = 0.606). Therefore, although the risk scores successfully separated patients into groups with significantly different survival outcomes, its ability to quantitatively discriminate individual patient outcomes was limited in the analysed cohort.

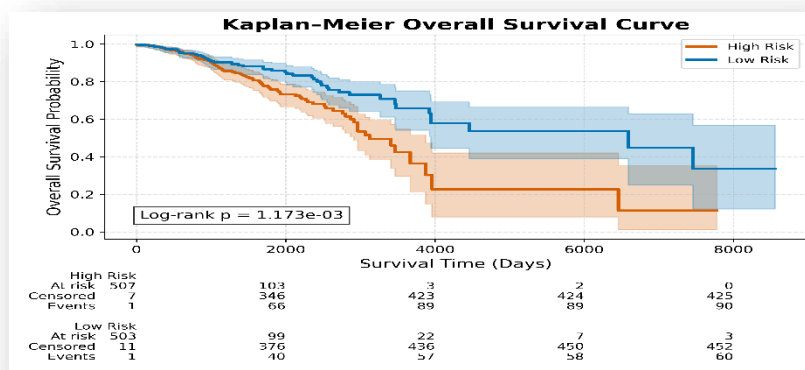


Figure 5.10. Kaplan–Meier overall survival analysis according to prognostic risk group.

Kaplan–Meier curves comparing overall survival between high- and low-risk patients defined according to the median prognostic risk score. Shaded regions represent **95% confidence intervals**. The difference between survival distributions was assessed using the log-rank test ($p = 1.173 \times 10^{-3}$). The number-at-risk table shows patients remaining at risk, censored observations, and events at the indicated time points.

5.11 Time-Dependent ROC Analysis

Time-dependent receiver operating characteristic (ROC) analysis was performed to assess the discriminative performance of the three-gene prognostic risk score at **1, 3, and 5 years**. The corresponding areas under the ROC curves (AUCs) were **0.527, 0.549, and 0.606**, respectively.

The risk score showed **limited discrimination at 1 year and 3 years**, with AUC values close to 0.5. Discrimination was higher at 5 years (**AUC = 0.606**), although the overall performance remained modest. These findings indicate that the three-gene risk score provided **limited-to-modest discrimination of overall survival at the evaluated time points** in the analysed TCGA-BRCA cohort.

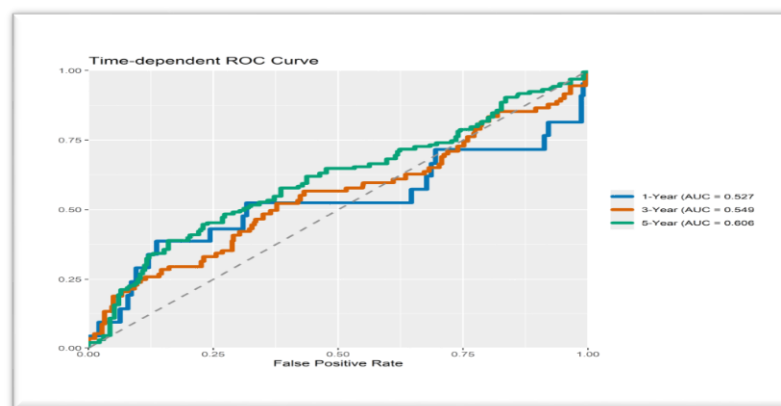


Figure 5.11. Time-dependent ROC curves for the three-gene prognostic risk score at 1, 3, and 5 years.

The ROC curves illustrate the discriminative performance of the risk score at the respective survival time points.

The AUC values were 0.527, 0.549, and 0.606 at 1, 3, and 5 years, respectively.

5.12 Concordance Index and Internal Validation

The discriminative performance of the three-gene prognostic model was evaluated using the concordance index (C-index). The model achieved an **apparent C-index of 0.5658** in the complete TCGA-BRCA cohort. To assess potential optimism in the apparent performance, **1,000 bootstrap resamples** were performed. The estimated mean optimism was **0.0012**, resulting in an **optimism-corrected C-index of 0.5646**. The bootstrap distribution yielded a **95% confidence interval of 0.5014–0.6342**.

Five-fold cross-validation was additionally performed, with the model fitted independently within each training fold and predictions generated for the corresponding held-out patients. The resulting **five-fold cross-validated C-index was 0.5706**.

Overall, the C-index estimates were relatively consistent across the apparent, optimism-corrected, and cross-validated analyses, indicating **modest discriminative performance** of the three-gene prognostic model in the analysed cohort.

Table 5.7. Internal validation of the three-gene prognostic model

Validation measure	Estimate	Interpretation
Apparent C-index	0.5658	Model performance in the full TCGA-BRCA cohort
Optimism-corrected C-index	0.5646	Bootstrap-adjusted estimate after correcting for optimism
Five-fold cross-validated C-index	0.5706	Out-of-fold discrimination across five validation folds
Bootstrap 95% confidence interval	0.5014–0.6342	Uncertainty range from 1,000 bootstrap resamples

5.13 Differential Expression Analysis

Differential expression analysis was performed between the high-risk ($n = 515$) and low-risk ($n = 515$) groups to identify genes associated with differences in the molecular profiles of the two risk groups. A total of 47,718 genes were evaluated. Gene-expression differences were assessed using Welch's two-sample t -test, and P values were adjusted for multiple testing using the Benjamini–Hochberg false discovery rate (FDR) procedure. Genes satisfying both $\text{FDR} < 0.05$ and $|\log_2\text{FC}| > 1$ were considered significantly differentially expressed.

Among the 47,718 genes analysed, 1,283 genes met the predefined differential-expression criteria, including 1,153 upregulated genes and 130 downregulated genes in the high-risk group relative to the low-risk group. Following gene-type annotation, 18,958 protein-coding genes were identified among the analysed genes. Of these, 913 protein-coding genes met the predefined significance criteria, comprising 830 upregulated and 83 downregulated genes.

The volcano plot demonstrated a greater number of significantly upregulated genes than downregulated genes in the high-risk group. The heatmap of the top 50 DEGs further illustrated differences in gene-expression patterns across patients ordered according to their prognostic risk score. Overall, the expression profiles showed a marked shift between the low- and high-risk portions of the cohort, consistent with the differential-expression analysis.

XG, which was subsequently retained as one of the three genes in the final prognostic signature, was also among the differentially expressed genes represented in the top-gene expression analysis. This provides complementary evidence that XG expression differs between the risk-stratified groups. However, differential expression and prognostic association represent distinct statistical observations and do not by themselves establish a causal role of XG in breast cancer progression or survival.

Table 5.8. Summary of differential expression analysis

Differential-expression	Number of genes	criterion
Total genes analysed	47,718	Genes tested between high- and low-risk groups
Significant DEGs	1,283	FDR < 0.05 and $ \log_2\text{FC} > 1$
Upregulated DEGs in high-risk group	1,153	Higher expression in high-risk patients
Downregulated DEGs in high-risk group	130	Lower expression in high-risk patients
Protein-coding genes analysed	18,958	Genes annotated as protein-coding
Significant protein-coding DEGs	913	Protein-coding DEGs meeting the significance criteria
Upregulated protein-coding DEGs	830	Protein-coding genes higher in high-risk patients
Downregulated protein-coding DEGs	83	Protein-coding genes lower in high-risk patients

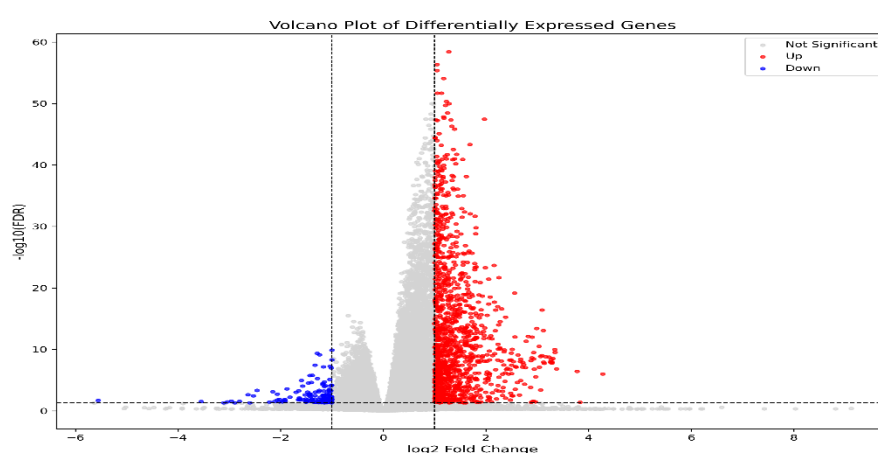


Figure 5.13. Volcano plot of differentially expressed genes between high- and low-risk groups.

The volcano plot shows the relationship between log2 fold change and $-\log_{10}(\text{FDR})$. Red points represent significantly upregulated genes, blue points represent significantly downregulated genes, and grey points represent genes that did not satisfy the predefined significance criteria.

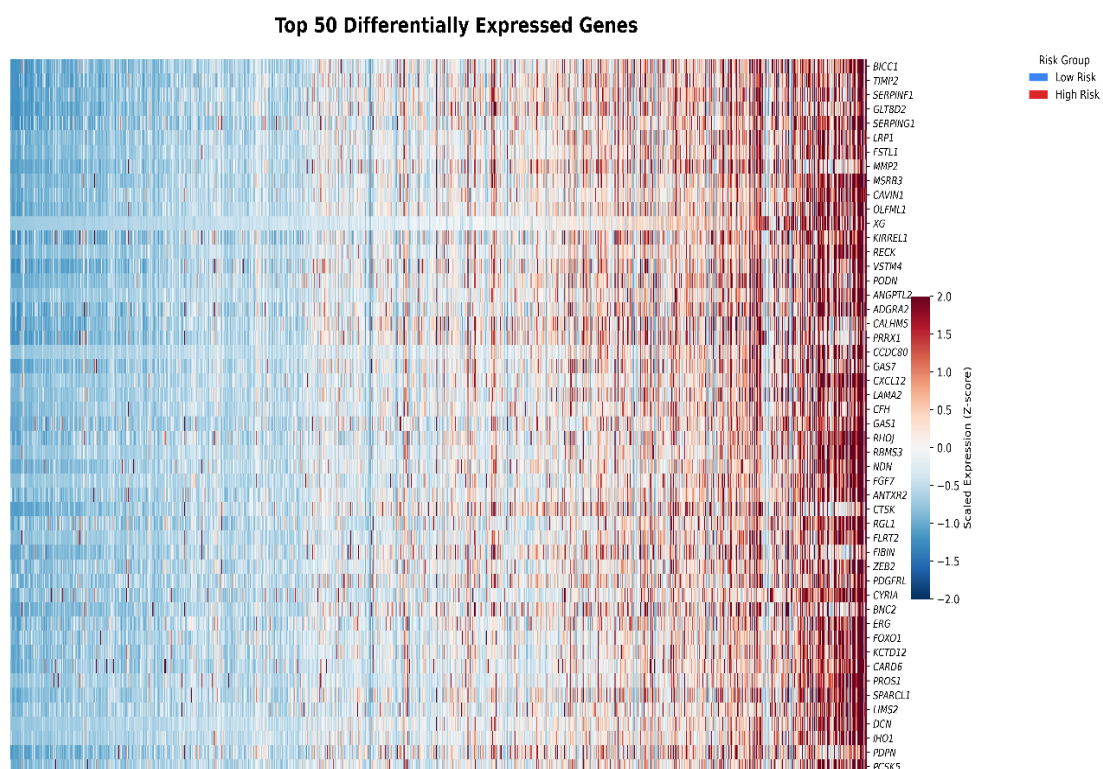


Figure 5.14. Heatmap of the top 50 differentially expressed genes.

The heatmap illustrates standardized expression patterns of the top 50 DEGs across the TCGA-BRCA cohort. Patients were ordered according to increasing prognostic risk score from low risk to high risk. Blue and red represent relatively lower and higher standardized expression, respectively.

5.14 GO enrichment analysis

GO enrichment analysis of the **913 significant protein-coding DEGs** identified **250 significantly enriched Biological Process (BP)**, **27 Cellular Component (CC)**, and **43 Molecular Function (MF) terms** at adjusted $P < 0.05$. The BP analysis showed a strong enrichment of processes related to **extracellular-matrix organization and remodelling**, with *Extracellular Matrix Organization* representing the most prominent term (40/176 genes; adjusted $P = 5.62 \times 10^{-14}$). Other closely related processes included *Extracellular Matrix Disassembly*, *Extracellular Structure Organization*, and *External Encapsulating Structure Organization*, indicating substantial involvement of extracellular and tissue-structural programs in the transcriptional differences between the risk groups.

Additional enriched BP categories involved **vascular and cellular regulatory processes**, including *Regulation of Angiogenesis*, *Negative Regulation of Blood Vessel Morphogenesis*, *Regulation of Cell Migration*, *Positive Regulation of Cell Population Proliferation*, *Positive Regulation of Cell Differentiation*, and *Positive Regulation of the MAPK Cascade*. These findings collectively indicate that the genes differentially expressed between the high- and low-risk groups participate in biological programs involving **tissue remodelling, vascular regulation, cellular movement, proliferation, differentiation, and intracellular signalling**.

The BP analysis also identified enrichment of **metabolic and hormone-responsive processes**, including *Long-Chain Fatty Acid Transport*, *Retinoid Metabolic Process*, *Cellular Response to Hormone Stimulus*, *Cellular Response to Peptide Hormone Stimulus*, and *Metal Ion Transport*. Thus, the molecular differences between risk groups were not restricted to structural processes but also involved metabolic regulation and cellular responses to extracellular stimuli.

The **Cellular Component** analysis provided additional structural context. The strongest enrichment was observed for the **collagen-containing extracellular matrix**, followed by *Intermediate Filament*, *Keratin Filament*, *Cell-Cell Junction*, *Caveola*, and several ion-channel-associated complexes, including calcium-, sodium-, and potassium-channel complexes. This pattern is consistent with changes in extracellular architecture, cytoskeletal organization, cell–cell interactions, and membrane-associated functions across the risk groups.

The **Molecular Function** analysis further supported these findings, with enrichment of **serine-type peptidase activity, serine-type endopeptidase activity, metalloendopeptidase activity, chemokine receptor binding, chemokine activity, voltage-gated ion-channel activities, fibroblast growth factor receptor binding, and several dehydrogenase activities**. The enrichment of peptidase and metallopeptidase functions is particularly consistent with the BP enrichment of extracellular-matrix disassembly and remodelling, whereas chemokine-related functions provide evidence for altered extracellular signalling.

Overall, the GO enrichment results indicate that the transcriptional differences associated with the prognostic risk groups are characterized primarily by coordinated alterations in extracellular-matrix organization and remodelling, vascular and cell-migration processes, proliferative and differentiation programs, signalling responses, and metabolic/hormonal functions. The concordance between BP, CC, and MF categories strengthens the biological

interpretation of the DEG analysis, although GO enrichment alone demonstrates functional association rather than establishing a causal mechanism for breast-cancer survival.

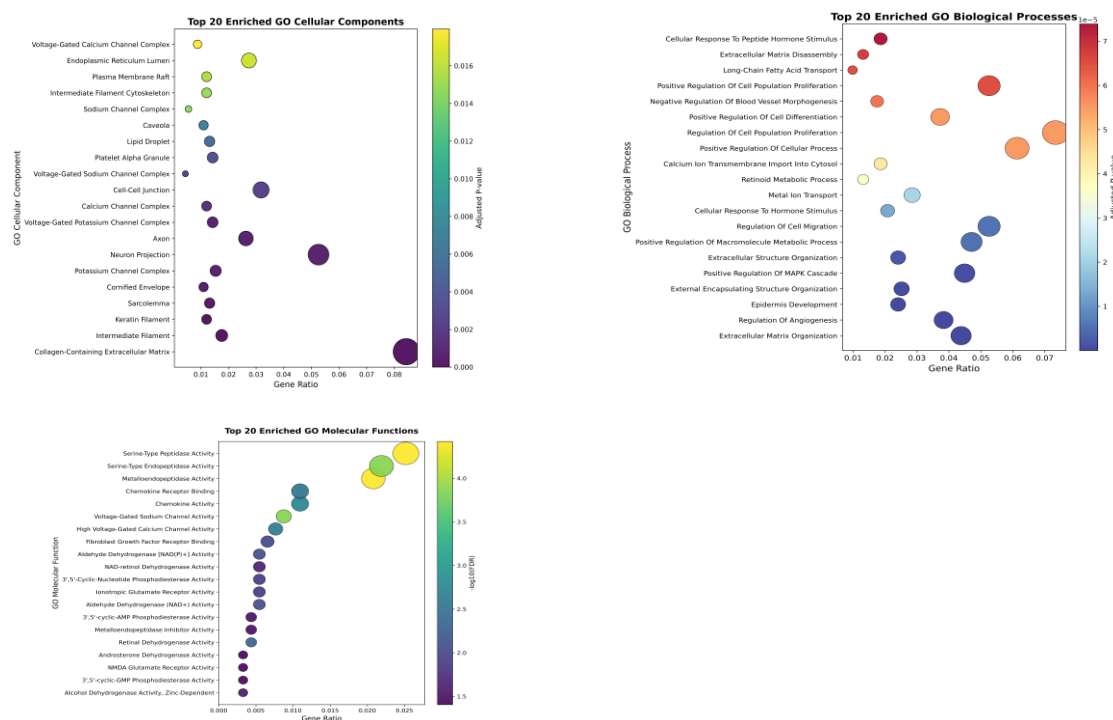


Figure 5.15. Gene Ontology enrichment analysis of significant protein-coding differentially expressed genes. Dot plots showing the top enriched (A) **Biological Process**, (B) **Cellular Component**, and (C) **Molecular Function** terms. Dot size represents the number of genes associated with each GO term, while colour represents the **Benjamini–Hochberg-adjusted *P*-value**.

5.15 KEGG Pathway Enrichment Analysis

KEGG pathway enrichment analysis was performed using the **913 significant protein-coding DEGs** identified between the high- and low-risk groups. A total of **261 enriched pathways** were identified, of which **28 were significant after Benjamini–Hochberg** multiple-testing correction (adjusted $P < 0.05$).

The major enriched pathways included the PPAR signalling pathway, calcium signalling pathway, PI3K-Akt signalling pathway, cytokine–cytokine receptor interaction, ECM-receptor interaction, retinol metabolism, ABC transporters, and complement and coagulation cascades. These pathways represent functions related to cellular signalling, extracellular-matrix interaction, cytokine-mediated communication, and metabolic regulation.

The enrichment of the **ECM-receptor interaction** pathway was consistent with the strong extracellular-matrix-related enrichment observed in the GO analysis, while enrichment of the **PI3K-Akt and calcium signalling pathways** provided additional evidence for altered cellular signalling between the prognostic risk groups. The enrichment of PPAR and retinol metabolism pathways further indicated involvement of **metabolic processes**.

Overall, KEGG analysis provided pathway-level support for the biological themes identified through GO enrichment, particularly **extracellular-matrix organization, cellular signalling, cell–cell communication, and metabolic regulation**. These findings provide functional context for the molecular differences between the high- and low-risk groups but do not establish causal relationships with breast-cancer survival.

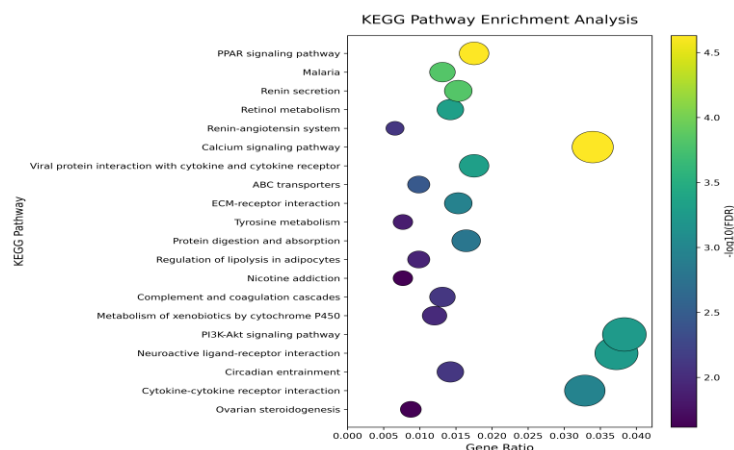


Figure 5.16. KEGG pathway enrichment analysis of significant protein-coding differentially expressed genes. Dot plot showing the top enriched KEGG pathways. Dot size represents gene count, colour represents the adjusted P -value, and the x-axis represents Gene Ratio.

5.17 Gene Set Enrichment Analysis (GSEA)

Gene Set Enrichment Analysis (GSEA) was performed using a ranked list of **18,955 protein-coding genes** to identify coordinated differences in gene-expression programs between the high- and low-risk groups. Significant enrichment was observed for **E2F Targets** (NES = -2.324 , FDR < 0.001), **epithelial–mesenchymal transition** (NES = 1.845 , FDR < 0.001), **TNF- α signaling via NF- κ B** (NES = 1.696 , FDR = 2.11×10^{-4}), **inflammatory response** (NES = 1.653 , FDR = 8.43×10^{-4}), and **angiogenesis** (NES = 1.650 , FDR = 7.22×10^{-4}).

Other significant pathways included **hypoxia**, **TGF- β signaling**, **IL-6/JAK/STAT3 signaling**, **adipogenesis**, and **fatty-acid metabolism**.

Overall, GSEA indicated coordinated differences in gene-expression programs related to **cellular regulation**, **EMT**, **inflammation**, **angiogenesis**, **signaling**, and **metabolism**. These findings were broadly consistent with the GO and KEGG analyses, particularly their enrichment of **cell migration**, **extracellular-matrix organization**, **angiogenesis**, **cytokine signaling**, and **metabolic processes**. The results provide additional functional context for the molecular differences between the prognostic risk groups but do not establish causality.

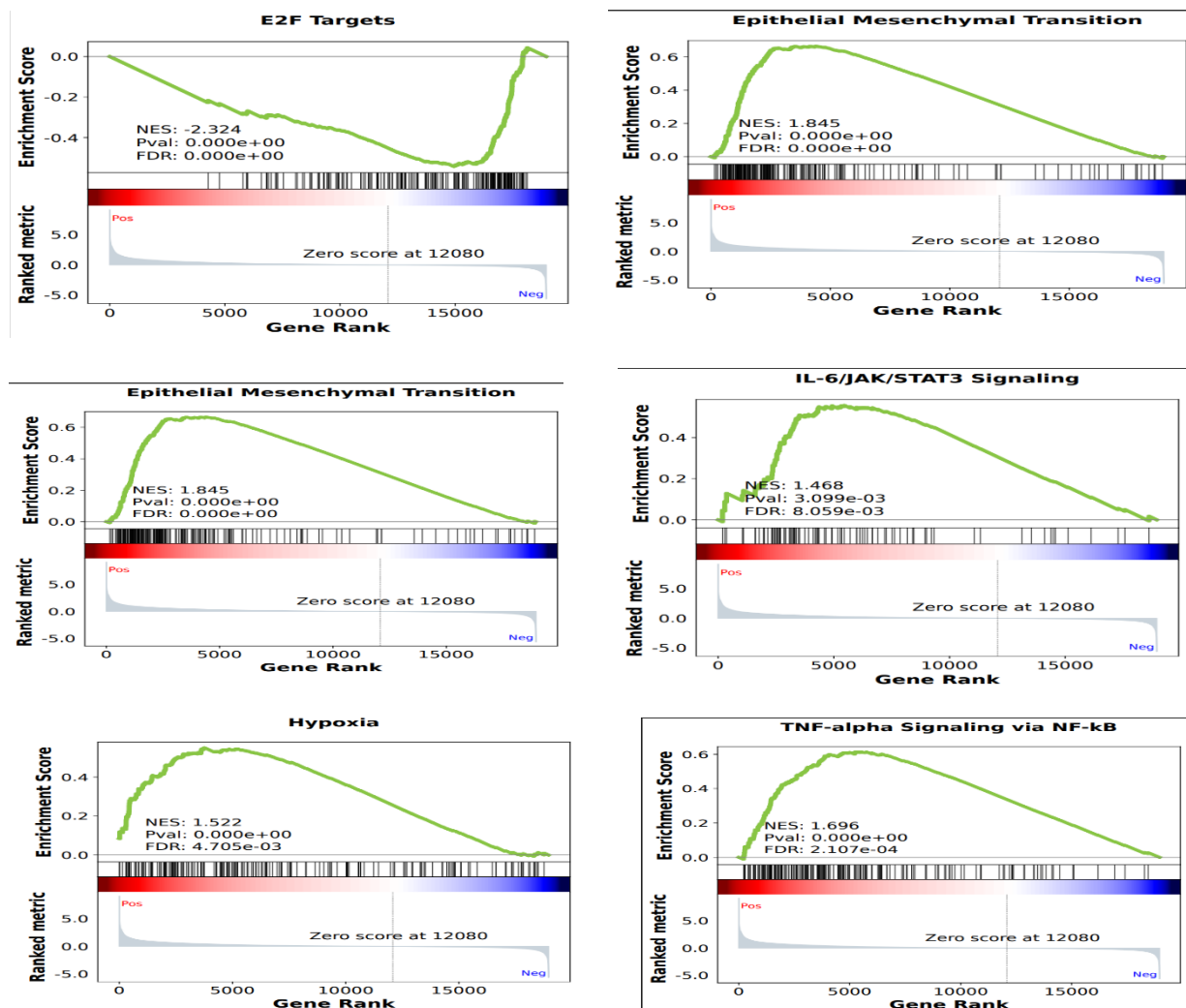


Figure 5.17. Gene Set Enrichment Analysis of selected pathways.

GSEA demonstrated significant enrichment of **E2F Targets**, **epithelial-mesenchymal transition**, **IL-6/JAK/STAT3 signalling**, **hypoxia**, and **TNF- α signalling via NF- κ B** in the ranked gene-expression profile. The plots show the enrichment score across the ranked gene list, with the corresponding normalized enrichment score (NES), nominal *P*-value, and false discovery rate (FDR) indicated within each panel.

5.18 PPI Network and Hub-Gene Analysis

5.18.1 STRING PPI Network Analysis

The 913 significant protein-coding differentially expressed genes were subjected to protein–protein interaction (PPI) analysis using STRING. The resulting interaction network was imported into Cytoscape for network analysis and identification of highly connected genes. Network topology showed several highly connected nodes, with IL6 exhibiting the highest degree (98) and the highest betweenness centrality (0.252). Other highly connected genes included GNG2 (degree = 58), PPARG (54), FGF2 (48), TLR4 (46), CXCL12 (44), LEP (42), IGF1 (42), DCN (42), and CD36 (40).

The network contained genes associated with inflammatory signaling, growth-factor-related processes, extracellular-matrix organization, and metabolic regulation, providing a network-level representation of the molecular differences identified between the prognostic risk groups.

5.18.2 MCC-Based Hub-Gene Analysis

The Cytoscape network was further analysed using the **Maximal Clique Centrality (MCC)** method to identify highly central nodes. The top-ranked hub genes were **PPARG (MCC score = 1700), ADIPOQ (1600), LIPE (1505), IL6 (1226), CXCL12 (1211), PLIN1 (973), CXCL2 (944), FOS (917), EGR1 (909), and FOSB (890)**.

Among these, **PPARG, ADIPOQ, LIPE, and PLIN1** represent prominent metabolic/lipid-associated components of the network, whereas **IL6, CXCL12, and CXCL2** represent inflammatory and chemokine-associated components. **FOS, EGR1, and FOSB** form a transcriptional regulatory component of the identified network.

Overall, the PPI analysis identified a network characterized by interconnected **metabolic, inflammatory, chemokine, and transcriptional regulatory components**. The hub-gene analysis therefore provided additional network-level support for the biological processes identified through the preceding GO, KEGG, and GSEA analyses.

Figure 5.18. Protein–protein interaction network of selected hub genes identified by MCC analysis. The network illustrates interactions among the selected hub genes and their associated interacting proteins. The network highlights interconnected inflammatory/chemokine, metabolic, and transcriptional component.

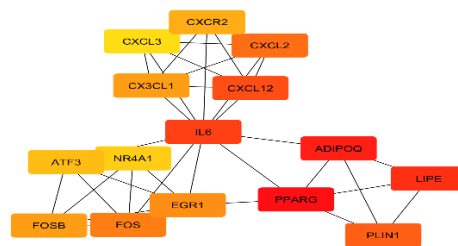


Table 5.9 Top 10 hub genes identified by MCC analysis

Rank	Gene	MCC Score
1	PPARG	1700
2	ADIPOQ	1600
3	LIPE	1505
4	IL6	1226
5	CXCL12	1211
6	PLIN1	973
7	CXCL2	944
8	FOS	917
9	EGR1	909
10	FOSB	890

5.16 Integrated Biological Interpretation

The differential expression and functional enrichment analyses revealed a consistent biological pattern between the high- and low-risk groups. The **913 significant protein-coding DEGs** were associated with **extracellular-matrix organization and remodelling, cell migration, angiogenesis, inflammatory signalling, and metabolic regulation**. GO and KEGG analyses further supported enrichment of extracellular-matrix and cell–matrix interactions, cytokine-related pathways, **PI3K-Akt and calcium signalling**, and metabolic pathways.

GSEA provided additional support for these findings, with significant enrichment of **epithelial–mesenchymal transition, angiogenesis, TNF- α /NF- κ B signalling, IL-6/JAK/STAT3 signalling, hypoxia, TGF- β signalling, adipogenesis, and fatty-acid metabolism**. These pathways are biologically relevant to breast-cancer progression, particularly through interactions among the tumour microenvironment, inflammatory signalling, vascular remodelling, cellular plasticity, and metabolism.

PPI analysis further identified highly connected genes including **IL6, PPARG, FGF2, CXCL12, TLR4, and CD36**, while MCC analysis identified **PPARG, ADIPOQ, LIPE, IL6, CXCL12, PLIN1, CXCL2, FOS, EGR1, and FOSB** as the top network hubs.

The presence of inflammatory/chemokine and metabolic hub genes is consistent with the functional enrichment results and provides additional network-level support for the identified biological themes.

Overall, the integrated analyses suggest that the prognostic risk groups are characterized by coordinated differences involving **extracellular-matrix remodelling, inflammatory and cytokine signalling, angiogenesis, cellular plasticity, and metabolic regulation**. These findings provide biological context for the prognostic signature; however, the enrichment and PPI analyses demonstrate **association rather than causation**, and the identified hub genes should not be interpreted as independently validated prognostic or therapeutic targets without external and experimental validation.

5.20 Prognostic Model Performance

The three-gene prognostic signature comprising **ABCB5, XG, and AC011483.2** was used to calculate individual risk scores for the 1,030 TCGA-BRCA patients. Using the median risk score (**−0.0772**) as the cutoff, the cohort was divided into **515 high-risk and 515 low-risk patients**. Kaplan–Meier analysis demonstrated significantly different overall survival between the two groups (**log-rank $P = 1.173 \times 10^{-3}$**), indicating an association between the calculated risk score and survival.

However, the discrimination metrics indicated limited predictive performance. The apparent C-index was **0.5658**, while bootstrap optimism correction produced a C-index of **0.5646** (95% CI: **0.5014–0.6342**). Five-fold cross-validation produced a similar C-index of **0.5706**. Time-dependent AUC values were **0.527, 0.549, and 0.606** at 1, 3, and 5 years, respectively. Thus, although the risk groups showed statistically significant survival separation, the model demonstrated **weak discrimination for individual survival prediction**, with performance only slightly above chance overall. The close agreement between bootstrap-corrected and cross-validated C-index values suggests that the observed performance was relatively consistent under the applied internal validation procedures.

5.21 Molecular and Network Characterization of the Risk Groups

Differential expression analysis between the high- and low-risk groups identified **913 significant protein-coding DEGs**, comprising **830 upregulated and 83 downregulated genes**. Functional enrichment analyses demonstrated prominent involvement of **extracellular-matrix organization and remodelling, cell migration, angiogenesis, inflammatory signalling, cellular signalling, and metabolic processes**.

GSEA further identified significant enrichment of **epithelial–mesenchymal transition, TNF- α /NF- κ B signalling, IL-6/JAK/STAT3 signalling, angiogenesis, hypoxia, TGF- β signalling, adipogenesis, and fatty-acid metabolism**, supporting the biological themes identified by GO and KEGG analyses.

STRING–Cytoscape analysis identified highly connected genes including **IL6, GNG2, PPARG, FGF2, TLR4, CXCL12, LEP, IGF1, DCN, and CD36**. MCC analysis further identified **PPARG, ADIPOQ, LIPE, IL6, CXCL12, PLIN1, CXCL2, FOS, EGR1, and FOSB** as the top-ranked hub candidates.

The enrichment and network findings collectively indicate involvement of **inflammatory signalling, extracellular-matrix interactions, angiogenesis, cellular signalling, and metabolic programs** in the molecular differences between the prognostic risk groups.

5.22 Association Between Risk Groups and Clinicopathological Characteristics

The association between the **low-risk (n = 515)** and **high-risk (n = 515)** groups was evaluated using available clinicopathological characteristics. The median age was similar between the low- and high-risk groups (**59.2 vs. 58.8 years, respectively; $P = 0.0864$**). Sex distribution also did not differ significantly between the groups ($P = 0.0640$).

A significant difference was observed in **race distribution** ($P < 0.05$). The proportion of White patients was higher in the high-risk group (**74.8%**) than in the low-risk group (**64.1%**), whereas Asian patients represented **4.1%** of the high-risk group compared with **7.6%** of the low-risk group. The proportion of patients with race not reported was also lower in the high-risk group (**5.8%**) than in the low-risk group (**10.7%**).

AJCC stage was significantly associated with risk group ($P = 0.0301$). Stage II represented **59.2%** of the low-risk group compared with **51.5%** of the high-risk group, whereas Stage III accounted for **17.5%** and **24.5%**, respectively. No significant difference was observed for **T stage** ($P = 0.8774$), with T2 being the predominant category in both groups. In contrast, **N stage** showed a significant association with risk group ($P = 0.0091$), with N2 occurring more frequently in the high-risk group (**13.6%**) than in the low-risk group (**7.8%**).

M stage did not differ significantly between the groups ($P = 0.9515$), with M0 being predominant in both low-risk (**80.8%**) and high-risk (**80.0%**) groups. However, **vital status differed significantly** ($P = 0.0104$), with deaths occurring in **17.5%** of the high-risk group compared with **11.7%** of the low-risk group.

Overall, the risk groups showed significant differences in **race, AJCC stage, N stage, and vital status**, whereas **age, sex, T stage, and M stage** were not significantly different between the groups. These findings indicate that the prognostic risk classification was associated with selected clinicopathological characteristics, particularly **disease stage, lymph-node status, and survival status**.

Table 5.10. Clinical characteristics of the TCGA-BRCA cohort according to risk group

Characteristic	Overall, n (%)	Low risk, n (%)	High risk, n (%)	P-value
Age, median (IQR), years	59.0 (49.7–67.8)	59.2 (50.4–68.5)	58.8 (49.3–66.4)	0.0864
Sex				0.0640
Male	11 (1.1)	9 (1.7)	2 (0.4)	
Female	1,019 (98.9)	506 (98.3)	513 (99.6)	
Race				0.0003
White	715 (69.4)	330 (64.1)	385 (74.8)	
Black/African American	169 (16.4)	91 (17.7)	78 (15.1)	

Asian	60 (5.8)	39 (7.6)	21 (4.1)	
Race not reported	85 (8.3)	55 (10.7)	30 (5.8)	
AJCC pathological stage				0.0301
Stage I	185 (18.0)	92 (17.9)	93 (18.1)	
Stage II	570 (55.3)	305 (59.2)	265 (51.5)	
Stage III	216 (21.0)	90 (17.5)	126 (24.5)	
Stage unknown	7 (0.7)	4 (0.8)	3 (0.6)	
T stage				0.8774
T1	254 (24.7)	124 (24.1)	130 (25.2)	
T2	580 (56.3)	296 (57.5)	284 (55.1)	
T3	118 (11.5)	55 (10.7)	63 (12.2)	
T4	33 (3.2)	18 (3.5)	15 (2.9)	
N stage				0.0091
N0	477 (46.3)	254 (49.3)	223 (43.3)	
N1	321 (31.2)	163 (31.7)	158 (30.7)	
N2	110 (10.7)	40 (7.8)	70 (13.6)	
N3	64 (6.2)	27 (5.2)	37 (7.2)	
M stage				0.9515
M0	828 (80.4)	416 (80.8)	412 (80.0)	
M1	19 (1.8)	9 (1.7)	10 (1.9)	
MX	140 (13.6)	69 (13.4)	71 (13.8)	
Vital status				0.0104
Alive	880 (85.4)	455 (88.3)	425 (82.5)	
Dead	150 (14.6)	60 (11.7)	90 (17.5)	

6. Discussion

The present study used transcriptomic and survival data from the **TCGA-BRCA cohort** to investigate molecular differences associated with prognostic risk in breast cancer. A prognostic risk score was used to stratify **1,030 patients into low- and high-risk groups**, followed by differential expression, functional enrichment, pathway, GSEA, protein–protein interaction, hub-gene, and clinicopathological analyses.

The risk stratification demonstrated a significant difference in overall survival between the low- and high-risk groups, indicating that the derived risk score was associated with patient survival. However, the discrimination performance of the model was relatively weak, with a concordance index of approximately **0.57** and time-dependent AUC values ranging from approximately **0.53 to 0.61**. Therefore, although the risk scores successfully separated the

cohort into groups with different survival outcomes, its current performance should be considered primarily as a **research-level prognostic stratification tool rather than a clinically validated predictive model**.

A major finding was the identification of **ABCB5** as one of the genes associated with the prognostic risk groups. ABCB5 expression was substantially higher in the high-risk group, with median expression values of **23 compared with 5** in the low-risk group. The difference was highly significant based on the Mann–Whitney U test ($P = 1.99 \times 10^{-45}$), with a calculated $\log_2$ fold-change of **3.09**. ABCB5 therefore represents a strong molecular feature distinguishing the two risk groups in this dataset. However, the observed association should not be interpreted as evidence that ABCB5 directly causes poor survival, and further independent and experimental validation would be required.

The differential expression analysis identified **913 significant protein-coding genes**, demonstrating substantial transcriptional differences between the risk groups. Functional enrichment analysis indicated that these genes were particularly associated with **extracellular-matrix organization and remodelling, cell migration, angiogenesis, cellular signalling, and metabolic processes**. The enrichment of extracellular-matrix-related processes was supported across GO Biological Process, Cellular Component, and Molecular Function categories, suggesting coordinated changes involving tissue architecture and cell–matrix interactions.

KEGG analysis further identified enrichment of pathways including **PPAR signalling, calcium signalling, PI3K-Akt signalling, cytokine–cytokine receptor interaction, ECM-receptor interaction, complement and coagulation cascades, retinol metabolism, and renin-related pathways**. These pathways represent diverse biological processes rather than a single mechanism, indicating that the transcriptional differences between the risk groups may involve interactions between **stromal, metabolic, inflammatory, and signalling programs**.

GSEA provided complementary evidence because it evaluated the complete ranked protein-coding gene list rather than only the genes passing the differential-expression threshold. Significant enrichment was observed for **epithelial–mesenchymal transition, coagulation, TNF- α /NF- κ B signalling, KRAS signalling, inflammatory response, angiogenesis, hypoxia, TGF- β signalling, IL-6/JAK/STAT3 signalling, fatty-acid metabolism, complement, and apoptosis**. These findings are broadly consistent with the GO and KEGG results and suggest that the prognostic groups differ in coordinated biological programs related to **tumour-associated signalling, inflammation, extracellular remodelling, vascular processes, and metabolism**.

The PPI and Cytoscape analyses identified highly connected genes including **PPARG, ADIPOQ, LIPE, IL6, CXCL12, PLIN1, CXCL2, FOS, EGR1, and FOSB**. The presence of IL6, CXCL12, FOS, EGR1, and FOSB provides additional support for the inflammatory and signalling themes observed in the enrichment analyses. However, the high connectivity of **PPARG, ADIPOQ, LIPE, and PLIN1** should be interpreted cautiously because these genes are strongly associated with adipocyte biology. Since TCGA-BRCA represents bulk tumour

tissue, their expression may partly reflect differences in **adipose or stromal tissue composition** rather than tumour-cell-specific regulation.

The clinicopathological analysis showed significant associations between the risk groups and **race, AJCC stage, N stage, and vital status**, whereas age, sex, T stage, and M stage were not significantly different. Stage III disease was more frequent in the high-risk group (**24.5% vs. 17.5%**), and N2 disease was also more frequent (**13.6% vs. 7.8%**). The proportion of deceased patients was higher in the high-risk group (**17.5% vs. 11.7%**). These observations provide clinical context for the molecular differences identified between the risk groups, although they do not establish that the molecular signature independently determines disease progression.

Overall, the results indicate that the prognostic risk groups are characterized by coordinated transcriptional differences involving **extracellular-matrix remodelling, inflammatory and cytokine signalling, angiogenesis, cellular signalling, metabolism, and tumour-associated biological processes**. ABCB5 emerged as a particularly prominent candidate gene because of its strong differential expression between the risk groups. Nevertheless, the findings remain **associative and computational**, and independent cohort validation and experimental studies are required before these observations can be translated into clinical or mechanistic conclusions.

7. Conclusion

The present study investigated transcriptomic and prognostic characteristics of breast cancer using the **TCGA-BRCA cohort**. Risk-based stratification identified significant survival differences between low- and high-risk patients. Comparative transcriptomic analysis identified **913 protein-coding differentially expressed genes**, which were enriched in biological processes involving extracellular-matrix organisation, cellular communication, angiogenesis, inflammation, signalling, and metabolism.

GSEA further supported the involvement of **EMT, inflammatory signalling, angiogenesis, hypoxia, TGF- β , IL-6/JAK/STAT3, KRAS, and metabolic pathways**. PPI and Cytoscape analysis identified several highly connected genes, including **PPARG, ADIPOQ, LIPE, IL6, CXCL12, and FOS-family genes**.

Among the candidate molecular features, **ABCB5 showed markedly higher expression in the high-risk group** and represents a potential prognostic candidate for further investigation. The risk groups were also associated with selected clinicopathological characteristics, particularly **AJCC stage, lymph-node stage, and vital status**.

Taken together, this study provides a **computational framework for identifying molecular features and biological pathways associated with breast-cancer prognosis**. The findings generate candidate genes and pathways for subsequent validation but do not establish causal mechanisms or clinical utility.

8. Limitations

1. The study was based primarily on **retrospective TCGA-BRCA data**, and therefore the findings are subject to the limitations of retrospective datasets.

2. The prognostic model showed **limited discrimination**, with a C-index of approximately 0.57. Therefore, the model should not currently be considered suitable for individual clinical prediction.
3. The analysis was primarily performed using **bulk transcriptomic data**, which cannot distinguish tumour-cell-specific expression from contributions of stromal, immune, adipose, and other cell populations.
4. The identified associations, including the association of **ABCB5** with risk status, do not demonstrate causality.
5. The PPI and hub-gene analyses are dependent on the available interaction database and network-construction criteria. High network connectivity does not necessarily indicate biological or prognostic importance.
6. The study requires **independent external validation** to determine whether the observed prognostic associations are reproducible in other breast-cancer cohorts.
7. Experimental validation was beyond the scope of the internship study; therefore, the biological functions of the identified candidate genes and pathways remain to be confirmed.

9. Future Perspectives

Future studies should focus on **independent validation of the prognostic findings in external breast-cancer cohorts** and evaluation of whether the identified molecular features remain significant after adjustment for established clinicopathological variables.

Attention should be given to **ABCB5**, which showed a strong association with the high-risk group. Experimental studies could investigate its expression and functional role in breast-cancer cells, including its potential relationship with **cellular signalling, tumour-associated inflammation, and treatment response**.

Further investigation of the tumour microenvironment using methods such as **immune-cell and stromal-cell deconvolution** could help determine whether the observed extracellular-matrix, inflammatory, and adipocyte-associated signatures originate primarily from tumour cells or differences in tissue composition.

The identified pathways, including **EMT, PI3K-Akt, inflammatory signalling, TGF- β , IL-6/JAK/STAT3, angiogenesis, and metabolic pathways**, could also be investigated experimentally to determine whether they contribute to the biological differences associated with the prognostic groups.

Ultimately, integration of **genomic, transcriptomic, clinical, and tumour-microenvironment data**, followed by experimental validation, may improve the robustness and biological interpretation of prognostic signatures in breast cancer.

10. References

TCGA-BRCA and breast cancer

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Important: Only include **DESeq2** or **limma** if you actually used that software/package in your analysis. From the workflow you've shown me, I don't want to falsely claim one of them.

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Software and Bioinformatics Tools Used

1. **Python 3.11:** Used as the primary programming environment for transcriptomic data processing, statistical analysis, survival analysis, and visualization.
2. **R:** Used for statistical and bioinformatics analyses, including data processing, statistical testing, survival analysis, and visualization where applicable.
3. **Pandas:** Used for importing, cleaning, filtering, merging, and manipulating gene-expression and clinical datasets.
4. **NumPy:** Used for numerical calculations and array-based data processing.
5. **SciPy:** Used for statistical analyses, including the **Mann–Whitney U test** for comparison of gene-expression levels between risk groups.
6. **Matplotlib:** Used to generate graphical representations, including **survival-related plots, GO enrichment plots, pathway plots, and clinicopathological distribution figures.**
7. **GSEAPy:** Used to perform **Gene Set Enrichment Analysis (GSEA)** using the ranked list of protein-coding genes and to identify significantly enriched biological gene sets.
8. **Gene Ontology (GO):** Used for functional enrichment analysis of significant protein-coding differentially expressed genes to identify enriched **Biological Process, Cellular Component, and Molecular Function** categories.
9. **KEGG:** Used for **pathway enrichment analysis** to identify biological and signalling pathways associated with the differentially expressed genes.

10. **STRING:** Used to construct **protein–protein interaction (PPI) networks** and identify functional associations among the significant protein-coding genes.
11. **Cytoscape:** Used for **visualization and network analysis** of the STRING-derived PPI network.
12. **cytoHubba:** Used as a Cytoscape plugin for **hub-gene identification and ranking**, with the **Maximal Clique Centrality (MCC)** method used to identify highly connected candidate hub genes.
13. **TCGA/GDC:** The **TCGA-BRCA dataset** obtained through the Genomic Data Commons was used as the primary source of breast-cancer transcriptomic and clinical data.